

Minimal Computing Cosmology 2.3.2

A Cumulative Integration of Minimal Execution, a Matter-Occupied Reality, the Asymmetric Unification of Four Forces, and a Finite Boundaryless Twist Universe

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Preservation Declaration

Minimal Computing Cosmology 2.3.2 does not select one earlier version and replace the others. It uses WRRRA Core 1.0 and the complete execution skeleton of MCC 2.0 as its foundation, then cumulatively integrates the selective microscopic reinforcement of 2.1, the twist stress universe of 2.2, and the standard physics module interface of 2.3.1.

$$MCC_{2.3.2} = MCC_{2.0}^{core} \oplus MCC_{2.1}^{micro} \oplus MCC_{2.2}^{twist} \oplus MCC_{2.3.1}^{modules}$$

The symbol $\oplus$ denotes a non-substitutive combination that preserves ownership. The Standard Model does not delete the common carrier. String theory does not replace minimal computation or phenotype. Twist does not displace the matter-occupied Reality or the asymmetric unification of the four forces. Conversely, restoring the earlier achievements does not delete the global and local twist construction or its calculations.

Abstract

Minimal Computing Cosmology describes the universe not as a stored list of completed future events but as a finite execution system in which minimal input and reusable laws update the present and produce one Reality and Record. Our universe need not be privileged or optimal. Minimal computation is not computation reduced without limit; it is the nearest sufficient computation, including reserve capacity, that avoids fatal failure under changing and extreme conditions. Luck enters the executed history as differences in encountered initial position, perturbation, and path among lawful events, rather than as a separate force.

Version 2.3.2 restores and integrates the central results of the research lineage. It connects LAW SOURCE EXECUTION RENDERER REALITY and RECORD in one genealogy. It constructs a constraint-preserving boundary map in which one opening input updates both the geometric and matter ledgers, and it connects that opening to kination, energy transfer, radiation domination, and the hot Big Bang without an additional primordial radiation register. It establishes an asymmetric unification of the four forces from the difference between the gravitational owner map, which

reads total stress energy, and the three gauge owner maps, which read selective currents. It places fifteen chiral channels under one common carrier while retaining the failure of democratic rank-one coupling, the serial-rotor construction, and the gauge-commutant limit. It defines matter as a jointly owned phenotype of mass, charge, spin, location, history, and record. It also separates lawful antimatter from actual occupancy, allowing a long-lived Reality composed of matter while preserving the conjugate antimatter sector in LAW. Quantum possibility is connected to one event and an accumulated record through lawful support, instruments, recovery, and long-time stability.

The twist universe of 2.2 remains a full component of this architecture. It defines global holonomy and bidirectional twist in a finite boundaryless Actual, and decomposes one twist state into a global mean and local deviation. The global component is rendered as the FLRW distance, redshift, and background-acceleration phenotype. The local component is rendered as the stress and mass phenotype associated with dark-matter gravity. The native $J(Y)$ and $K(Q)$ dynamics of 2.2 and the CDM-equivalence module of 2.3.1 do not replace one another. The former is the native branch in which twist directly produces $T_{\mu\nu}^\Theta$; the latter is the equivalence branch in which $T_{\mu\nu}^\Theta \equiv T_{\mu\nu}^{CDM}$ within a declared domain so that validated calculations are inherited. The two sources are never added twice.

The Standard Model, general relativity, FLRW, and the currently validated cosmological calculation are retained as Reality modules. String theory is selected only when microscopic material for quantum gravity, anomaly closure, chiral zero modes, or the common carrier is required. The center of 2.3.2 is therefore not a defense against established physics. It is the preservation of the results already obtained by the minimal execution architecture and twist cosmology while validated calculations are connected under explicit and non-duplicating ownership.

Keywords

Minimal Computing Cosmology WRRRA minimal computation common carrier asymmetric force unification matter phenotype matter occupied

Reality antimatter Actual Reality Record constraint preserving opening hot
Big Bang finite boundaryless universe twist stress dark matter mass
phenotype FLRW Lambda CDM string theory

Core Contributions

Ledger — Contribution · Preserved result in 2.3.2 · Status

Contribution: Single execution architecture

Preserved result in 2.3.2: One genealogy from LAW through SOURCE
execution rendering reality and record

Status: Core architecture

Contribution: Minimal cosmic opening

Preserved result in 2.3.2: Failure of the bare kick and a constraint-
preserving canonical boundary map

Status: Constructed exact result

Contribution: Hot Big Bang connection

Preserved result in 2.3.2: No Extra Radiation Register and radiation
takeover for $Q_r \geq 0$

Status: Conditional dynamical closure

Contribution: Asymmetric unification of four forces

Preserved result in 2.3.2: Nonreducibility of total $T_{\mu\nu}$ ownership to
selective J_a ownership

Status: Theoretical result

Contribution: Fifteen-channel common carrier

Preserved result in 2.3.2: Rank-one failure serial rotor full abstract span
and commutant limit

Status: Construction and no-go boundary

Contribution: Matter as phenotype

Preserved result in 2.3.2: Joint ownership of mass charge spin location
and record

Status: Theoretical architecture

Contribution: Matter-occupied actual world

Preserved result in 2.3.2: Separation of lawful antimatter from occupied matter Reality

Status: Ontological result with open numerical asymmetry

Contribution: Quantum possibility to one record

Preserved result in 2.3.2: Lawful support instrument event history and recovery

Status: Constructed execution logic

Contribution: Finite execution benchmark

Preserved result in 2.3.2: Positive self-adjoint Jacobi visible port recovery gap and thermal ledger

Status: Finite exact benchmark

Contribution: Finite boundaryless twist

Preserved result in 2.3.2: Global holonomy bidirectional twist and nonprivileged gluing

Status: Topological model

Contribution: Dark mass phenotype

Preserved result in 2.3.2: Local stress $T_{\mu\nu}^{\Theta}$ and effective gravitating mass

Status: Calculable model branch

Contribution: Twist interpretation of expansion

Preserved result in 2.3.2: Correspondence between the FLRW scale and global connection updates

Status: Interpretive correspondence

Contribution: Modular standard physics

Preserved result in 2.3.2: Nonduplicating inheritance of the Standard Model GR FLRW and Lambda CDM

Status: Modular integration

Version Preservation Rules

Ledger — Source · Scope preserved in 2.3.2 · Nonreplacement rule

Source: WRRR Core 1.0

Scope preserved in 2.3.2: Minimal computation common carrier phenotype present update ownership boundary record and ledger

Nonreplacement rule: A cosmological profile cannot silently revise the Core

Source: MCC 2.0

Scope preserved in 2.3.2: All eighteen chapters core equations constructions failures and results

Nonreplacement rule: Standard physics modules do not become the skeleton

Source: MCC 2.1

Scope preserved in 2.3.2: Selective string microphysics under the three invariants

Nonreplacement rule: A string state does not replace the common carrier or phenotype

Source: MCC 2.2

Scope preserved in 2.3.2: First twist finite boundarylessness bidirectional twist stress galaxies lensing and early universe

Nonreplacement rule: CDM inheritance does not delete native twist dynamics

Source: MCC 2.3.1

Scope preserved in 2.3.2: Ownership Standard Model GR FLRW CDM interface structure calculation and revision order

Nonreplacement rule: An unfinished interface does not precede the original achievements

Parts I through IV Complete Skeleton of Minimal Computing Cosmology 2.0

Part I The Minimal Execution Structure of the Universe

Big Bang, Dimensions, Laws, and Minimal Input

Chapter 1 One Universe One Execution Architecture

The first claim of Minimal Computing Cosmology is not the proposal of a new particle, but that physical phenomena usually described separately can be rearranged as one process of generation, execution, and recording.

Modern physics is extraordinarily precise, yet it answers a single overarching question in different languages. Classical mechanics describes the motion of bodies, relativity describes spacetime and causal structure, and quantum mechanics describes the amplitudes and probabilities of possible events. But no single ledger explains how laws begin one universe, how matter and forces arise together within it, and why one recorded reality continues among many possibilities.

To fill this gap, Minimal Computing Cosmology proposes WRRRA, the Reality-Renderer Architecture. WRRRA does not discard the successful equations of existing theories. Instead, it places each equation into a single execution order according to what it owns and at which stage it acts. Unification here does not mean flattening different theories into one equation; it means connecting the process by which one reality is produced while preserving the distinct responsibilities of the component theories.

Statement

The Book in One Sentence A fixed LAW defines the space of possibilities and relational rules; a minimal SOURCE opens one history; common transport and selective renderers jointly implement spacetime, matter, and forces; and quantum events accumulate as one actual record within that execution.

LAW → SOURCE → EXECUTION → RENDERER → REALITY → RECORD (1.1)

Here, LAW owns the physical laws, constants, state space, symmetries, conservation laws, and boundary conditions. SOURCE is not a completed blueprint of the universe but the minimal boundary input that opens a prepared possibility. EXECUTION updates the present state into the next present. RENDERER turns internal degrees of freedom into a shared observable representation. REALITY is the physically realized visible sector of the entire Actual, and RECORD is the irreversibly accumulated history of events.

With these distinctions, the Big Bang, dimensions, the four forces, mass, vacuum, antimatter, quantum probability, and a single history are no longer unrelated chapters but different positions in one flow. The purpose of this book is not to rewrite tens of thousands of known facts in new notation. It is to show, with minimal structure, why those facts can be executed together in one universe without contradiction.

Ledger — Established Theory · Primary Question · Position within WRRR

Established Theory: Classical Mechanics

Primary Question: How do realized objects move?

Position within WRRR: Motion of stable macroscopic phenotypes

Established Theory: Relativity

Primary Question: How do spacetime and causal structure operate?

Position within WRRR: Common causal geometry and gravitational ledger

Established Theory: Quantum Mechanics

Primary Question: What amplitudes and probabilities do possible events have?

Position within WRRR: Quantum refinement of the allowed event space

Established Theory: WRRR

Primary Question: How do possibilities become one Reality and one Record?

Position within WRRR: Higher-level source-to-reality execution architecture

Chapter 2 Three Forms of Minimality

In this theory, ‘minimal’ does not merely mean that an equation is short. It means jointly reducing the number of dimensions that must be physically realized, the number of independent inputs, the amount of transport structure, and the scope that must be recomputed at every moment.

The first is minimal dimensionality. To explain the observable universe, WRRRA does not require as a premise a large number of physical extra dimensions that cannot be directly visualized. The physical stage of our universe closes in three spatial directions and one temporal ordering. If additional dimensions are introduced, one must first distinguish whether they are dimensions of an internal representation space, a state space, an arithmetic registry, or actual spacetime dimensions. A large number of internal degrees of freedom does not by itself imply a high-dimensional physical space.

Statement

Academic Meaning of Minimality Independent prediction is one axis for evaluating a theory. But the capacity to explain many already known facts together, without contradiction and with fewer dimensions, inputs, owners, and transport rules, is also a central criterion in theory choice.

The second is minimal structure. If charge, mass, spin, position, the three gauge forces, and gravity are each produced in separate factories and then attached afterward, the number of coupling rules proliferates. WRRRA reuses a common state, common transport, common boundaries, and selective representations. Diversity is obtained not by multiplying independent origins, but by allowing the same structure to be read through different ledgers.

The third is minimal computation. The universe need not pre-store every future event or reread its laws from outside at every moment. Fixed laws and the current state generate the next present; observable differences are output only at the boundaries where they are needed; and only events that actually occur are recorded.

$$C_{\text{total}} = L_{\text{LAW}} + L_{\text{SOURCE}} + C_{\text{read}} + C_{\text{evolution}} + C_{\text{render}} + C_{\text{record}} \quad (2.1)$$

Thus, minimization targets the total explanatory cost rather than a single count of operations. If one model requires more physical dimensions, more independent initial values, more separate carriers, and more post hoc corrections to explain the same observations, then it has a larger structural cost over the same explanatory domain.

$$A^* = \text{Pareto-min} \{D_phys, L_SOURCE, N_owner, N_carrier, C_update \mid A \models \Omega_observed\} \quad (2.2)$$

Rather than rank all models by an arbitrary single number, this framework uses Pareto minimality. A structure is a minimal candidate when reducing one item while preserving observational compatibility necessarily increases another. WRRRA is a candidate that achieves a broad explanatory range with 3+1 dimensions, one opening, reused common transport, and local updating.

Chapter 3 The Minimal SOURCE and the Opening of the Universe

The Big Bang is not an explosion of a completed data set for the universe. It is a constraint-preserving boundary transition in which a possibility dormant under fixed laws enters one actual history.

If the first cause of the universe contains a list of particles, all future events, and every value at every point in space, the explanation never really begins, because one must again explain who created that information, where it was stored, and how it is read. Minimal Computing Cosmology replaces the first cause as an information list with a single operation acting on a structured possibility.

$$P = \{H_potential, \Omega, P_Q, H_WRRRA, G, A\}, [\Omega, P_Q] = i\hbar \quad (3.1)$$

$$\rho_dorm = F(H_potential, G, A, C), \text{ free coefficients} = 0 \quad (3.2)$$

The dormant state is not nothingness but a possibility prepared by law. However, if arbitrary initial coefficients are hidden in this state, the claim that there is only one SOURCE collapses. Therefore, uniqueness of the invariant state, residual degeneracy, block weights, and superselection rules must all be exposed either as part of LAW or as explicit boundary data.

$$U_S = \exp(i\eta\Omega/\hbar), U_S^\dagger P_\Omega U_S = P_\Omega + \eta \quad (3.3)$$

The parameter eta is not a particle or a list of energy values. It is a primordial boundary input that starts one execution by shifting the momentum of a prepared opening degree of freedom. Matter and events that appear afterward are not items read out of SOURCE; they are generated through the interaction of LAW with the evolving state.

Restricting the system to flat FLRW geometry and a homogeneous scalar chi, a bare kick that changes only the momentum of chi violates the Hamiltonian constraint. This provides a clear counterexample to a simple primordial-impulse model. By contrast, the following boundary generator shifts the scalar and gravitational momenta together and preserves the constraint.

$$B_\eta(a, \chi) = \eta \chi - \sqrt{6} \bar{M}_P |\eta| \ln(a/a^*) + \text{constant} \quad (3.4)$$

$$p\chi^+ = \eta, p\alpha^+ = -\sqrt{6} \bar{M}_P |\eta|/a^*, H^* = |\eta|/(\sqrt{6} \bar{M}_P a^{*3}) \quad (3.5)$$

The central point of this result is not that ‘the Big Bang has been proven,’ but something more specific. The opening of the universe cannot be an arbitrary impulse acting only on the matter sector; it must be a canonical boundary map that changes the geometry and the energy ledger together. The Big Bang is not an explosion inside space, but a transition in which space, clock, and transport begin to operate together.

3.1 From Opening to the Hot Big Bang

As the simplest effective branch after the constraint-preserving opening, consider an interval in which the kinetic energy of chi dominates. The kination scaling ρ_χ proportional to a^{-6} is a BASELINE result of standard cosmology. What WRRR newly connects is this branch directly to the opening with $p_\chi^+ = \eta$, while refusing to insert a radiation component as a new initial value in SOURCE.

$$\rho_\chi^* \simeq \eta^2/(2a^{*6}), \rho_\chi(a) = \rho_\chi^* (a^*/a)^6 \quad (3.6)$$

Radiation is not supplied as an independent primordial register. Suppose that the opening sector and interactions already contained in LAW own a nonnegative energy-transfer function Q_r . Then the total energy ledger can

be conserved while the evolution is written as the following pair of equations.

$$\dot{\rho}_\chi + 6H\rho_\chi = -Q_r, \dot{\rho}_r + 4H\rho_r = Q_r, Q_r \geq 0 \quad (3.7)$$

In particular, radiation can be generated during EXECUTION even if $\rho_r(t^*) = 0$ at the instant of opening.

$$\rho_r(t) = a(t)^{-4} \int_{[t^*]}^t a(t')^4 Q_r(t') dt' \quad (3.8)$$

Therefore, if Q_r is positive on any finite interval, ρ_r becomes positive. No separate radiation seed needs to be inserted as SOURCE data. This is the No Extra Radiation Register condition.

Defining $R = \rho_r/\rho_\chi$, the transfer equations during the kination interval give the following relation.

$$\dot{R}/R = 2H + Q_r(1/\rho_r + 1/\rho_\chi) \geq 2H \quad (3.9)$$

Thus, while $Q_r \geq 0$, the ratio of radiation to opening-sector energy grows at least as rapidly as a^2 . Even after the transfer ends, ρ_r proportional to a^{-4} while ρ_χ proportional to a^{-6} , so sufficiently prolonged expansion leads to radiation domination. In the simple seed limit $Q_r = 0$, with $\epsilon_r = \rho_r^*/\rho_\chi^*$, one has $a_{RD}/a^* = \epsilon_r^{(-1/2)}$; continued transfer with $Q_r > 0$ does not delay this transition.

Accordingly, the minimal cosmological-history connection used in this book is: dormant $\rightarrow$ constraint-preserving opening $\rightarrow$ kination/energy transfer $\rightarrow$ radiation domination $\rightarrow$ hot Big Bang. The microscopic owner of Q_r and the actual transition temperature remain OPEN, but the need to add a separate radiation initial value to SOURCE in order to obtain a hot Big Bang is removed.

Chapter 4 Where Do Dimensions and Spacetime Come From

WRRRA does not treat dimensions as empty boxes given in advance. A dimension is a direction in which relations can be transmitted independently and maintained as stable records.

The dimension filter F selects, from the possible relations, the sector that passes locality, stability, information conservation, and boundary-record

conditions. The transmitted kernel and complementary sector are separated as follows.

$$K_{DF} = F^\dagger F, L_{DF} = I - K_{DF} \quad (4.1)$$

The advantage of this expression is that it does not prematurely divide visible space and invisible degrees of freedom into existence versus nonexistence. A degree of freedom may exist in the total Actual while failing to pass through the physical renderer. Dark sectors, internal representation spaces, and hidden correlations can therefore occupy different positions within the same distinction.

$$d_{\text{eff}}(t) = -2 \, d \log \text{Tr}(\exp(-tK_{DF})) / d \log t \quad (4.2)$$

The effective dimension can be read from spectral flow. This book does not claim to have derived three spatial dimensions uniquely and without input. The more precise achievement is to provide a minimal implementation path that preserves the observed 3+1-dimensional universe without requiring additional physical spacetime dimensions. The dimensionality of an internal 15-channel space or of Hilbert space is not an extra physical spatial dimension.

Time likewise need not be an external clock already flowing before the opening. Taking the Hamiltonian constraint as fundamental, conditional time can be obtained from correlations between a clock subsystem and a semiclassical branch after the opening.

$$\hat{C}|\Psi\rangle = 0, i\hbar \partial\tau |\psi_{\text{cond}}(\tau)\rangle = H_{\text{eff}}|\psi_{\text{cond}}(\tau)\rangle \quad (4.3)$$

A persistent space requires more than a metric of rank three at one instant. The metric at every time must remain uniformly comparable with a common reference, and reconstruction rules in overlapping local regions must agree with one another.

$$c \, g^* \leq g_k \leq C \, g^* \text{ (all } k), R_U|U \cap V \simeq R_V|U \cap V \quad (4.4)$$

Statement

The Precise Claim of Minimal Dimensionality WRRRA accommodates a 3+1-dimensional Reality, a large internal state space, and diverse particle representations without assuming eleven physical spacetime dimensions.

This is structural minimality, distinct from a proof of dimensional uniqueness.

Part II Asymmetric Unification of Forces and Matter

A Common Origin Does Not Erase Differences

Chapter 5 Why the Four Forces Cannot Be the Same Kind of Force

Unifying gravity with the three gauge forces does not mean flattening all four terms into a single gauge field. It means showing how four representations with different ownership rules can operate together within one execution of Reality.

Electromagnetism, the weak interaction, and the strong interaction selectively read particular representation currents carried by matter. Gravity, rather than selecting a particular charge, couples to the stress-energy of all physically realized degrees of freedom. This difference is not merely a difference in phenomenological strength; it means that the input ledgers are of different kinds.

$$G_{\mu\nu} + \Lambda g_{\mu\nu} = (8\pi G/c^4) T_{\mu\nu_total}, \nabla_\mu T_{\mu\nu_total} = 0 \quad (5.1)$$

$$D_\mu = \nabla_\mu^{\text{geo}} - ig_1 Y B_\mu - ig_2 T^a W_\mu^a - ig_3 t^A G_\mu^A \quad (5.2)$$

Let the owner map for gravity be $O_G(X)=T_{mn}[X]$, and the owner maps for the three gauge forces be $O_a(X)=J_a[X]$. A complete Standard-Model singlet X may satisfy $T_{mn}[X] \neq 0$ while $J_1[X]=J_2[X]=J_3[X]=0$. Gravity then reads X even though the three gauge forces do not.

$$\exists X : (J_1, J_2, J_3)[X] = 0 \wedge T_{\mu\nu}[X] \neq 0 \quad (5.3)$$

$$\text{therefore } O_G \neq f(O_1, O_2, O_3) \text{ for any current-only flattening } f \quad (5.4)$$

This counterexample shows that a flat unification reducing gravity completely to a selective gauge current of the same kind as the other three forces is not generally possible. Gravity reads the geometric ledger of the entire realization, while the three gauge forces selectively read internal representations within it. A common origin is therefore possible without requiring that all four forces be of the same type.

Statement

Asymmetric Unification Proposition Gravity = the common causal geometry of the entire realization. Electromagnetism, the weak interaction, and the strong interaction = selective internal gauge representations of the same local state. The four forces arise within one renderer, but they do not read the same ledger.

This structure also gives dark matter a natural place. Even a gauge singlet remains in the gravitational sector if it carries real energy, momentum, and pressure. Conversely, abstract information content or computational complexity alone is not a gravitational source. A contribution enters the gravitational ledger only when metric variation produces a nonzero T_{mn} .

WRRRA's unification is therefore not a 'weaker failure' than conventional grand unification; the goals are different. Grand unification seeks a common microscopic origin of gauge groups and coupling constants. WRRRA asks why gravity and gauge forces, within the same Reality, can simultaneously display universality and selectivity in different ways. These tasks may compete, but they should not be scored by an identical rubric.

Chapter 6 The Common Carrier How Diversity Emerges from One Structure

Minimal structure does not erase every difference. It must reuse common transport laws while preserving the charge, representation, and selection rules of each channel.

One generation of chiral matter carries multiple charges and representations. Treating the 15 directions as 15 unrelated origins increases both SOURCE complexity and transport rules. WRRRA instead places the 15 directions under one common continuum and one common transport operator, implementing their differences through internal labels and boundary readout.

$$D_C : \oplus_{i=1}^{15} H_i \rightarrow \oplus_{i=1}^{15} H_i, \text{rank}(D_C | \text{band})=15 \text{ (6.1)}$$

The common carrier is neither a particle nor a fifth force. It is a higher-level operator that allows different channels to share the same skeleton of local transport, kinematics, and boundary recording. Just as using the same

road does not give all vehicles the same destination or properties, commonality is compatible with distinguishability.

An early democratic rank-one coupling transmitted only one bright combination and left the other fourteen directions dark. This failure closed the shortcut that ‘symmetry alone implies unification.’ A serial rotor was then proposed in which a rotating rank-one port accumulates phase coding over time and opens the full span.

$$P(\varphi) = |\psi(\varphi)\rangle\langle\psi(\varphi)|, \text{rank } P(\varphi)=1, \text{rank span}\{\psi(\varphi_k)\}=15 \quad (6.2)$$

However, opening rank 15 in an abstract 15-dimensional space is not the same as realizing one gauge-covariant Standard-Model generation. The commutant of the Standard Model decomposes into five irreducible blocks, so a pure-state orbit that commutes with the gauge action cannot exceed rank five.

$$\text{End_GSM}(H_{15}) \leq 5 \times \text{rank}(\text{span orbit})$$

This result does not discard the common carrier. It clarifies that abstract carrier rank, gauge-invariant density support, and particle-antiparticle Fock production have different owners. The strength of WRRRA is not in forcing the number 15 to match everywhere, but in distinguishing which layer a given occurrence of 15 belongs to.

Chapter 7 Matter Is a Phenotype Not a List of Numbers

Obtaining mass, charge, spin, and position separately does not yet create one particle. A physical object exists when multiple attributes are jointly owned by the same execution and the same record.

The Standard Model is the precise phenotype grammar of observed matter. WRRRA does not relabel that success and claim it as its own. Instead, it asks at which layer of the universe’s execution architecture the Standard Model’s representations, couplings, and mass generation are owned.

$$m_f(\mu) = y_f(\mu) v(\mu) / \sqrt{2}, \det S_f^{-1}(p) | p^2 = m^2_{\text{pole}} = 0 \quad (7.1)$$

Charge and spin are representation properties, while mass is not a single number stored in SOURCE. It is an execution phenotype jointly produced by the Higgs-Yukawa structure, self-energy dressing, QCD binding, and stable poles, gaps, and thresholds. Therefore, an observed mass used as an

input must be labeled CALIBRATED; only a value obtained from upstream generative rules may be labeled DERIVED.

The phrase ‘one particle’ contains a hidden co-ownership condition. Mass, charge, spin, and position must be read together from the same local observable algebra, the same translation history, the same spectral pole, and the same event record. Placing four numbers from different calculations in one table is not the same thing as showing that one physical object exists.

Object $X \Leftrightarrow$ co-owned $\{m, q, s, x\}$ on $(A_local, history, pole, record)$ (7.2)

This notion of co-ownership extends beyond matter to complex systems in general. If it is not specified which functions, memories, resources, and outputs belong to the same object, explanation remains a list of correlations. WRRRA’s ownership concept is therefore not decorative terminology but a logical device for fixing ‘what is a state of what.’

Ledger — Information Grade · Meaning · Example

Information Grade: INPUT

Meaning: Explicitly supplied by the law or the model

Example: $c, \hbar$, observed coupling constants

Information Grade: DERIVED

Meaning: Calculated from inputs and execution rules

Example: Time evolution of a specified model

Information Grade: CALIBRATED

Meaning: Value fitted to observation

Example: Fixing a scale using the electron mass

Information Grade: OBSERVED

Meaning: Value read from external data

Example: Present cosmic composition; observed events

Information Grade: OPEN EVENT

Meaning: An outcome newly recorded during execution

Example: An individual detection outcome

Information Grade: OPEN

Meaning: Owner or value not yet determined

Example: Upstream derivation of generations and Yukawa couplings

Chapter 8 An Actual World Composed of Matter

The fact that the laws symmetrically permit matter and antimatter does not mean that the actual universe must continuously occupy both sectors in equal amounts.

Three layers are mixed together in the matter-antimatter problem: first, the symmetry permitted by the equations; second, pair production in particular processes; and third, the occupation state of the actual universe over long times. Once these three layers are separated, a reality composed of matter need not conflict with the antimatter channels permitted by the laws.

Lawful sectors = $H_{\text{matter}} \oplus H_{\text{antimatter}}$, Occupied Reality(t) $\subset H_{\text{matter}}$ (8.1)

This proposition does not say that antimatter fails to exist mathematically. Antimatter is a conjugate sector allowed by LAW and can be physically output together with matter when the required conditions are present, such as collisions, external fields, or nonstationary boundaries. But in an ordinary long-lived reality without such events, the antimatter sector has no obligation to remain continuously occupied.

WRRRA's physical vacuum follows the same distinction. Vacuum is not the zero vector of the Actual, nor a state in which every field is literally zero. It is a phenotype in which, for a specified background, observer, and detector prescription, the normal real-particle output rate and net energy flux are zero.

$\Gamma_{\text{real-particle}}[\text{vacuum}; \text{background}, \text{detector}] = 0$ (8.2)

$\langle N \rangle = 0 \nRightarrow \{ \langle T_{\mu\nu} \rangle_{\text{ren}} = 0 \vee G_{\text{plus}}(x, x') = 0 \}$ (8.3)

Pair production is not a process in which 'existence continually emerges from nothing.' In a stable stationary background, the persistent production

rate of on-shell pairs is zero. Real output appears when particle definitions and energy exchange change, as in external fields, collisions, time-dependent boundaries, curvature, horizons, or accelerated observation. Pair annihilation likewise is not the deletion of existence but a conservative transfer of ownership from particle ports to other carrier modes such as photons.

matter + antimatter $\rightarrow$ carrier output, $\Delta\text{Ledge_total} = 0$ (8.4)

Statement

Matter-Only Reality Proposition The symmetry of the laws is a symmetry of the possibility space. Occupation of Reality is the result of events, boundaries, and stability. Therefore, antimatter can remain lawful while the persistent actual universe is composed of matter.

The problem of calculating the quantitative baryon asymmetry must be distinguished from this ontological possibility. The Sakharov conditions address the dynamics needed to explain why our universe acquired a particular asymmetry. WRR's prior conclusion is more basic: what is possible and what remains persistently actual need not be the same set.

8.1 Dynamics of a Matter Preponderant Reality

Here, 'matter preponderance' does not mean the cosmological matter-dominated era, in which ρ_m exceeds radiation or dark energy. In this chapter it means a matter-over-antimatter asymmetry in long-term occupation.

$$\Delta_M \equiv n_M - n_{Mbar} \quad (8.5)$$

When expansion, production, and annihilation are included together, symmetric annihilation terms cancel from the difference equation, and the actual occupation asymmetry can be tracked as follows.

$$d(a^3\Delta_M)/dt = a^3(S_M - S_{Mbar}) \quad (8.6)$$

After asymmetry generation ends, if $S_M - S_{Mbar} = 0$, then $a^3\Delta_M$ is conserved. In a state with $\Delta_M > 0$, if the symmetric matter-antimatter component annihilates sufficiently, n_{Mbar} becomes small and the remaining matter approaches the residual asymmetry. Thus, it is

compatible for LAW to allow antimatter channels while the actual long-lived Reality remains matter-preponderant.

The asymmetry itself, however, must not be inserted as a new primordial number. Δ_M^* must either be a function of η and LAW or arise from S_M-S_{Mbar} generated by existing interactions after the opening. This is the No Extra Asymmetry Register condition.

For a LAW and opening with exact CP symmetry, the boundary must not create an arbitrary matter direction.

$$F_B(\eta, \text{LAW_CP-symmetric}) = 0 \quad (8.7)$$

Therefore, to generate a real baryon asymmetry, the required baryon-number violation, C/CP asymmetry, and out-of-equilibrium execution must be explicitly owned by LAW or by interactions after the opening. The upstream calculation of the final numerical value $\eta_B = (n_B - n_{\text{Bbar}})/s$ remains OPEN, and this boundary is the same one applied to conventional baryogenesis.

Part III From Quantum Possibility to One Reality

Actual, Reality, Event, and an Open Single History

Chapter 9 Actual Reality Event

The total Actual and the visible Reality must be distinguished if information conservation, vacuum, measurement, and dark sectors are to be handled in one language.

Actual is the total dynamical state evolving under LAW. Reality is the sector jointly implemented through the physical observable algebra and the renderer. An Event is one newly distinguishable record added to that Reality. SOURCE is not the whole Actual, and Reality is not a screen that copies all of Actual.

$$\Omega_t \in S_{\text{Actual}} - R_t \rightarrow \rho_t \in S_{\text{Reality}} - I_a \rightarrow (a_t, \rho_{\{t+1\}}) \quad (9.1)$$

$$\rho_{\text{Reality}}(\tau) = P_{\text{vis}}[e^{(-iH\tau/\hbar)} U_S \rho_{\text{dorm}} U_S^\dagger e^{(+iH\tau/\hbar)}] \quad (9.2)$$

$$p(o, \tau) = \text{Tr}[E_o \rho_{\text{Reality}}(\tau)] \quad (9.3)$$

The renderer is not an arbitrary image generator. In a tensor-factor model it can be implemented as a CPTP channel such as a partial trace; under

restriction to an observable algebra it must preserve positivity, normalization, and the relevant conservation conditions. If several Actual states are compatible with the same Reality, the invisible differences remain in a compatibility fiber.

$$C(\rho) = \{\Omega : R(\Omega) = \rho\} \quad (9.4)$$

Information that does not appear in Reality has therefore not necessarily disappeared. It may remain in hidden or environmental correlations. This distinction allows black-hole information, vacuum correlations, quantum measurement, and macroscopic records to be discussed within one framework. Information is not simply a matter of ‘present or absent,’ but of who owns it and which observable algebra can read it.

The Actual-Reality distinction is not a declaration of two Platonic worlds. It aligns the distinctions already used in physics between total states, reduced states, observable algebras, and records with their ontological responsibilities. WRRRA’s additional claim is that these layers jointly produce one Reality within a single execution order.

Chapter 10 Fixed Laws and an Open Single History

There is no contradiction between laws being fixed and future events not yet being written. A closed conservation ledger and open events can coexist within one execution system.

Fixed laws constrain possible transitions, amplitudes, conserved quantities, and causal structure. But this does not mean that every future event must be stored as a classical list. WRRRA imposes the No Predetermined Event Register condition: neither LAW nor SOURCE contains a readable register that directly stores the future event sequence.

$$(a_1, a_2, \dots, a_\infty) \notin \text{readable registers of LAW or SOURCE} \quad (10.1)$$

A quantum instrument supplies the probabilities of possible outcomes and the conditional post-event states. Only one outcome is actually recorded, and it becomes part of the condition for the next present.

$$p(a | \rho) = \text{Tr}[I_a(\rho)], \quad \rho_a = I_a(\rho) / p(a | \rho) \quad (10.2)$$

$$\Psi_n \rightarrow \{a_1, \dots, a_k\}, \quad A_n \in \{a_1, \dots, a_k\}, \quad h_n = (h_{n-1}, A_n) \quad (10.3)$$

This structure does not require either a classical table of predetermined endings or the unrestricted actualization of every branch in a many-worlds picture. Possibilities are plural, but the actual record is singular. Outcomes not selected remain counterfactual possibilities, while the event that occurs is incorporated into the accumulated history. The future is conditionally open, and the past is closed in the record.

Finite execution experiments have constructed multiple sequential event histories from the same initial complex state and the same self-adjoint Hamiltonian while preserving unitarity and instrument completeness. This is an existence construction demonstrating the logical compatibility of fixed laws with open events.

$$\Sigma_a I_{a^\dagger}(Q)=Q, \quad ||U^\dagger U-I|| \approx 0, \quad \Sigma_a p(a|\rho)=1 \quad (10.4)$$

The information in a realized history can be written not as the size of a file stored from the beginning, but as the conditional surprise accumulated whenever an event occurs.

$$I_{\text{history}}(T)=\Sigma_t -\log_2 p(a_t|h_{\{t-1\}}) \quad (10.5)$$

Statement

The Information Economy of the Universe Short laws can generate a long history. The universe is not a device replaying a completed future; it is an execution system that updates the present under fixed laws and writes its own history.

Chapter 11 The Boundary Between Quantum Possibility and Macroscopic Reality

Quantum probability is not magic that can manufacture arbitrary events. There is first a space of events permitted by the laws, and amplitudes and Born weights operate only within that space.

WRRRA divides selective realization into two passes. The first pass constructs the support set Γ_{lawful} allowed by causality, energy, momentum, charge, gauge constraints, interaction history, and stable recording. The second pass assigns complex amplitudes and probabilities only to paths within that set.

$$\Gamma_{\text{lawful}} = C_LAW(p; \text{laws, constants, constraints, history}) \quad (11.1)$$

$$A[\gamma] = a[\gamma] \exp(iS[\gamma]/\hbar), K = \Sigma_{\{\gamma \in \Gamma_{\text{lawful}}\}} A[\gamma] \quad (11.2)$$

$$\text{supp } P_W\text{RRA} \subseteq \Gamma_{\text{lawful}} \quad (11.3)$$

This structure is not an after-the-fact prohibition switch designed to restrict quantum-mechanical calculations. In a completed model, Γ_{lawful} must be generated for both microscopic and macroscopic systems from the same states, environments, and transport rules. The central intuition is that probability does not manufacture physical possibilities from nothing; it assigns relative weights to events that are already lawful possibilities.

The difference between microscopic uncertainty and macroscopic definiteness is not divided by mass alone. When small event differences are redundantly stored through environmental records and interactions, macroscopic history becomes stable. A candidate scale combines the mass M , discontinuous displacement Δx , and record redundancy R_{rec} .

$$\Omega_W\text{RRA} = R_{\text{rec}} M c \Delta x / \hbar, \chi_{\text{est}} = \exp(-\Omega_W\text{RRA}) \quad (11.4)$$

This quantity is not yet a universal constant but a candidate boundary scale. The direction, however, is clear: the quantum-to-classical transition is not fundamentally a matter of ‘an observer looking,’ but of how redundantly and persistently the distinguishability of alternatives is stored in physical records.

Decoherence reduces interference between different record branches, but it is not the same operation as making only one branch actual. No deterministic selector based solely on linear unitary dynamics can send every superposition to one outcome while preserving linearity, norm, and Born weights simultaneously. Record formation and single-outcome selection must therefore be treated as distinct layers.

$$U(|\psi\rangle |E_0\rangle) = \Sigma_a c_a |a\rangle |E_a\rangle, \langle E_a | E_b \rangle \approx \delta_{ab} \quad (11.5)$$

Chapter 12 Conditions for One Reality to Remain the Same Reality

Outputting a Reality once is not enough to make a universe. Different rendering paths must yield the same physical content, that content must remain stable over long times, and local realities must glue together into one shared Reality.

Two rendering paths that begin at the same physical boundary and end at the same boundary must agree on protected observables even if their intermediate slices and gauge choices differ. This is renderer path independence.

$$R_{\{\gamma_1\}}(B_f, B_i) \simeq_{\text{gauge}} R_{\{\gamma_2\}}(B_f, B_i) \quad (12.1)$$

In canonical gravity, compositions of small deformations must close into another allowed deformation. If the constraint algebra does not close, different choices of slicing produce different physical realities.

$$\{H[N], H[M]\} = D[q^{ab}(N\partial_b M - M\partial_b N)] \quad (12.2)$$

A general quantum channel can contract information, so the mere fact that a map is CPTP does not preserve position or metric rank. What is required is not recovery of every microscopic state, but recovery on the physically protected observable algebra.

$$\text{Recovery} \circ \Phi = \text{id on } A_{\text{protected}} \quad (12.3)$$

$$R_1 \circ \Phi_{\{\gamma_1\}} \simeq R_2 \circ \Phi_{\{\gamma_2\}} \text{ on } A_{\text{protected}} \quad (12.4)$$

Recovery at one instant and long-term stability are also different requirements. Repeated disturbances must be summable, or there must be a noiseless observable algebra, or the recovery map must contract toward a code manifold.

$$d_{\{k+1\}} \leq \lambda d_k + \epsilon_k, 0 \leq \lambda < 1 \quad (12.5)$$

Uniform metric corridors and quasi-local gluing must be added to these conditions before multiple local observations can form one shared Reality. This family of conditions is an important independent contribution of WRRRA because it asks not only ‘what exists’ but what minimal conditions

allow the same existence to remain the same Reality across different observations and times.

Part IV The Meaning of Unification and a Computable Architecture

A Higher-Level Placement of Established Physics and a General Execution Grammar

Chapter 13 Placing Classical Relativistic and Quantum Physics in One Structure

A higher-level theory does not invalidate the precise calculations of lower-level theories. It preserves the questions and domains answered by each theory while explaining the execution order connecting them.

Classical mechanics describes motion extremely well after stable macroscopic records have formed. Relativity describes the common geometry and causal structure in which that motion occurs. Quantum mechanics describes the amplitudes, interference, and probabilities of allowed microscopic possibilities. WRRR asks how these three layers connect within one execution of the universe and become one actual record.

Motion → Causal Geometry → Quantum Possibility → Physical Realization
(13.1)

WRRR's higher-level role therefore does not mean merely reproducing all established equations as approximations to one new equation. More fundamentally, it connects the inputs and outputs of existing theories and assigns ownership by specifying what kind of object the output of one layer becomes for the next.

Ledger — Framework · Nature of Additional Assumptions · Mode of Unification

Framework: Parallel use of standard physics

Nature of Additional Assumptions: Independent equations and matching conditions for each domain

Mode of Unification: Selected by observational regime

Framework: String-theory family

Nature of Additional Assumptions: Higher dimensions, strings/branes, compactification

Mode of Unification: Unification of UV microstructure

Framework: Simulation cosmology

Nature of Additional Assumptions: Possibility of an external computer or implementer

Mode of Unification: Philosophical execution analogy

Framework: Minimal Computing Cosmology

Nature of Additional Assumptions: 3+1 physical dimensions, minimal SOURCE, renderer

Mode of Unification: Unification of ownership and execution order

This table alone cannot decide which framework is ultimately true, but it permits theoretical comparison. Minimal Computing Cosmology connects the Big Bang, matter, forces, quantum events, and a single history in one execution language without requiring physical space beyond the observed 3+1 dimensions or an external simulator. If the same explanatory range is achieved, structural economy is a substantive advantage.

In particular, WRRRA does not force gravity to become the same kind of entity as the three gauge forces. It preserves the difference between the geometry of the whole realization and selective internal representations while unifying them in one execution. A central feature of WRRRA is that unification must not be confused with identity.

Chapter 14 Turning the Declaration into a Finite Execution System

To avoid remaining a purely verbal framework, WRRRA constructed a finite benchmark in which a positive spectrum, a self-adjoint Hamiltonian, a visible port, a recovery gap, and an entropy ledger coexist.

$$H_N = H_{\text{Jacobi}} \otimes H_{\text{loc}} \otimes H_{\text{syn}} \quad (14.1)$$

The starting point is a positive generalized-Laguerre spectral density above a threshold s_0 . For $\alpha > -1$ it is normalizable, and the three-term

recurrence of the orthogonal polynomials determines a real symmetric Jacobi matrix.

$$\rho(s) = ((s-s_0)^\alpha e^{\{-(s-s_0)/\Lambda^2\}/(\Gamma(\alpha+1)\Lambda\{2\alpha+2\}}) \Theta(s-s_0) \quad (14.2)$$

$$(J_N)_{nn} = s_0 + \Lambda^2(2n + \alpha + 1), (J_N)_{n,n+1} = \Lambda^2 \sqrt{(n+1)(n+1+\alpha)} \quad (14.3)$$

If the first basis vector e_0 is chosen as the visible port, each eigenvalue weight is the square of its first component, so the weights are automatically nonnegative and sum to one. One collective coordinate reads N hidden spectral nodes. The Euclidean response and real-time amplitude arise from the same nodes and weights.

$$G_E(Q^2) = \int \rho(s)/(Q^2+s) ds = \sum_j w_j/(Q^2+s_j) \quad (14.4)$$

$$A_N(t) = \sum_j w_j e^{\{-is_j t\}} \quad (14.5)$$

For $N=2,4,8,16,32$, the maximum relative error over the specified Euclidean window decreased from 4.089×10^{-2} to 1.763×10^{-9} . A finite spectrum, however, retains long-time recurrence. This difference separates the ‘time window over which a finite system accurately imitates a continuum’ from ‘permanent irreversibility.’

In the recovery sector, an amplitude-damping-type Lindblad operator was introduced and the syndrome-transverse gap was calculated exactly.

$$L_{\text{rec}}(\rho) = \kappa[\sigma_- \rho \sigma_+ - \frac{1}{2}\{\sigma_+ \sigma_-, \rho\}], \Delta_{\text{corr}} = \kappa/2 \quad (14.6)$$

$$\Delta S_{\text{rec}} = -p \ln p - (1-p) \ln(1-p), Q_{\text{min}} = k_{\text{BT}} \Delta S_{\text{rec}} \quad (14.7)$$

The value of this benchmark is not that it replicates the entire universe. It is that positivity, self-adjointness, a common visible port, finite-window convergence, recovery, metric preservation, and a thermodynamic ledger - all required by WRRRA - can coexist without contradiction in one finite construction.

Chapter 15 Viewing Many Physical Phenomena with One Tool

WRRRA's explanatory power does not consist in inserting the same numbers into every phenomenon. It consists in reading different phenomena through the same system of questions: in which ledger and at which boundary does each phenomenon arise?

Ledger — Phenomenon · Core WRRRA Interpretation · Primary Owner

Phenomenon: Big Bang

Core WRRRA Interpretation: Constraint-preserving opening boundary transition

Primary Owner: SOURCE + geometry ledger

Phenomenon: Dark Matter

Core WRRRA Interpretation: A remainder invisible to gauge interactions but retained in T_{mn}

Primary Owner: gravitational ledger

Phenomenon: Black Hole

Core WRRRA Interpretation: Finite boundary information and the problem of internal transport and recording

Primary Owner: boundary + transport

Phenomenon: Temperature

Core WRRRA Interpretation: Sensitivity of the number of accessible states to energy

Primary Owner: coarse-grained state space

Phenomenon: Entropy

Core WRRRA Interpretation: Weighted multiplicity of histories indistinguishable from the current record

Primary Owner: observer/coarse-graining

Phenomenon: Vacuum

Core WRRRA Interpretation: Zero-real-particle-output phenotype on a stationary background

Primary Owner: detector + background

Phenomenon: Pair Production

Core WRRRA Interpretation: Conditional output opened by a driven configuration

Primary Owner: external work + field

Phenomenon: Pair Annihilation

Core WRRR Interpretation: Conservative transfer of ownership from particles to carrier modes

Primary Owner: conservation ledger

Phenomenon: Nuclear Decay

Core WRRR Interpretation: Multiple clocks associated with distinct barriers and transition rules

Primary Owner: channel-specific dynamics

Phenomenon: Superconductivity / Superfluidity

Core WRRR Interpretation: Collective phenotype of global phase stiffness, not merely local pairing

Primary Owner: boundary + phase relation

The important point in this table is not a claim that WRRR replaces the standard-physics calculation for every phenomenon. It separates problems previously blurred under the same words into owner, boundary, transport, and record, making clear which connections are shared and which dynamics are specific to each field. That is the precise meaning of ‘seeing them together.’

For example, temperature and entropy are not identified merely because both can be described using information language. The area-based capacity of a black hole, its current entropy, and the length of a generative description are also distinguished. Antimatter production is separated from persistent antimatter occupation, and quantum correlations from real-particle output. WRRR’s unification does not remove distinctions; it places them in the correct locations.

$$1/(k_{BT}) = \partial S_{acc}(E) / \partial E \quad (15.1)$$

$$S_{hist} = -k_B \sum_i p_i \ln p_i \quad (15.2)$$

These distinctions allow facts from different fields to be compared in a single ledger. In particular, they help determine whether explaining a phenomenon requires adding a new entity, whether a different output of an existing state is sufficient, whether an invisible sector carries real energy, or whether it is merely hidden computationally.

Chapter 16 The Minimal Execution Grammar Repeated Beyond Cosmology

The value of applying WRRRA to multiple fields is not that external applications automatically validate the cosmology. It is that the minimal questions required to construct complex reality recur across domains.

Cells, the origin of life, neural behavior, economics, and weather contain different physical components. Yet the need to distinguish independent states, relation networks, boundaries, residual history, update rules, and resource ledgers recurs. WRRRA expresses this common grammar as SOURCE, RELATION, BOUNDARY, RESIDUE, UPDATE, and LEDGER.

System = {SOURCE, RELATION, BOUNDARY, RESIDUE, UPDATE, LEDGER}
(16.1)

In life, a DNA sequence alone does not make an executable cell; translation machinery, energy, boundaries, catalysts, distribution, and intergenerational regeneration must close together. In neural behavior, the relevant question is not merely whether memory exists but which tasks require past history and what difference that memory makes to reward and safety. In weather and economics, one must test whether a weak historical signal actually outperforms a strong current-state baseline.

In these external applications, some models failed to beat their baselines. But the information contained in failure was preserved precisely through the common grammar. Rather than saying ‘there is no residue,’ one can say ‘the observed trajectory is not the owner of the residue’; rather than saying ‘memory is useless,’ one can say ‘the given task does not reward that memory.’ The location of failure can therefore be identified.

Statement

Value of a General-Purpose Tool A single equation need not output the numbers of every field. The common format instead determines who owns which information, what is closed by the present state, and where new events, environments, or boundaries are required.

This is stronger than a simple analogy. If the same structural questions operate independently on different data sets and repeatedly expose

incorrect ownership assignments, WRRR becomes a metatheory for comparing execution structures of reality. At the same time, it preserves logical independence by not converting success in a particular external domain into evidence for the cosmology itself.

Chapter 17 What Has Already Been Obtained

The value of a theory cannot be assessed solely through a list of what remains unfinished. Structural discoveries, exact constructions, NO-GO results, and quantities that remain to be calculated must be separated into different classes of achievement.

Ledger — Grade · What This Book Has Obtained

Grade: Core architecture

What This Book Has Obtained: Single flow of LAW-SOURCE-EXECUTION-RENDERER-REALITY-RECORD

Grade: Theoretical result

What This Book Has Obtained: Asymmetric unification produced by global T_{mn} ownership for gravity and selective ownership of three gauge currents

Grade: Ontological result

What This Book Has Obtained: Compatibility between lawful antimatter and an Actual Reality occupied by matter; conditional dynamical connection from $\Delta_M > 0$ and annihilation of the symmetric component to matter-preponderant occupation

Grade: Exact constructions

What This Book Has Obtained: Constraint-preserving boundary map in FLRW minisuperspace; finite Jacobi structure with positivity and self-adjointness

Grade: Boundary results

What This Book Has Obtained: Failure of a bare kick; obstruction to current-only gravity flattening; nonidentity of decoherence and single-outcome selection

Grade: Methodological result

What This Book Has Obtained: Provenance ledger for information ownership, input, derivation, calibration, observation, and events

Grade: Computational result

What This Book Has Obtained: Common visible port, fixed-window spectral convergence, recovery gap, and entropy ledger

Grade: Higher-level integration

What This Book Has Obtained: Staged placement of classical motion, relativistic geometry, quantum possibility, and single-Reality implementation

Grade: Cosmological-history connection

What This Book Has Obtained: Conditional route to the hot Big Bang after the constraint-preserving opening through radiation generated by Q_r and the different dilution rates of radiation and kination; No Extra Radiation Register

The remaining problems are also clear. A fully covariant 3+1-dimensional action, upstream derivations of Standard-Model generations, Yukawa couplings, mixing, and coupling constants, the E_{lift} into quantum Actual, the R_{phys} map fixing physical quantities and units, and the microscopic dynamics of the Born rule and single-outcome selection still require further calculation.

These unfinished items do not reduce the preceding achievements to zero. Just as the mathematical achievements of string theory's higher-dimensional unification do not disappear because vacuum selection and all low-energy phenomena remain unfinished, WRRR possesses theoretical structures and exact partial results that can be evaluated before independent predictions are completed.

Most important is not to insert established equations into WRRR and then call them WRRR achievements. Standard equations are labeled BASELINE; structures exactly constructed inside the model are labeled CONSTRUCTED/EXACT; structural connections are labeled THEORETICAL RESULT; values fitted to observation are labeled CALIBRATED; and

quantities whose owner is still unresolved are labeled OPEN. These distinctions are not apologetic caveats that weaken the theory, but boundaries that protect its actual achievements.

Chapter 18 What Still Needs to Be Completed

The next stage is not to diminish the value of the theory again, but to connect the central architecture already established to stronger mathematics and calculation.

The first task is to formalize minimality. A Pareto ledger must be completed that compares 3+1-dimensional physical space, SOURCE input length, the number of independent owners, the number of carriers, and update cost under the same observational contract. This turns the aesthetic claim that ‘nature should be simple’ into a computable comparison principle.

The second task is to extend the scope of the asymmetric-unification result. Starting from the gauge-singlet counterexample, one must show that the global geometric owner and selective current owners coexist consistently within a covariant action, anomaly freedom, symmetry breaking, and RG matching. This would give WRRRA’s unification of the four forces a sharper theoretical form.

The third task is to calculate the microscopic owners of the early-universe dynamical links already established. Q_r , which produces radiation from the opening, and S_B - S_{Bbar} , which produces the matter-antimatter asymmetry, must be closed as functions of the existing LAW and ϵ . Neither a radiation seed nor Δ_B may be added as a new SOURCE initial value. The ultimate targets are independent calculations of the transition temperature and $\epsilon_B = (n_B - n_{Bbar})/s$.

The fourth task is a single covariant execution expression. The FLRW opening, gauge matter, common carrier, quantum instrument, and renderer must be placed under a shared set of variables, symmetries, and constraint algebra.

$$S_{WRRRA}[g, \Phi_{open}, \Psi_{matter}, A_{gauge}, R] = \int d^4x \sqrt{-g} L_{total} + S_{boundary} \quad (18.1)$$

The fifth task is to compare explanatory compression directly. When conventional parallel theories, higher-dimensional unification models, and WRRRA explain the same set of phenomena, their required physical dimensions, independent inputs, separate carriers, calibrated values, and joining rules can be compared in an open ledger. This quantitatively evaluates explanatory economy independently of new predictions.

Statement

The Next Sentence of the Research The unfinished parts are not holes that negate the existence of the theory; they are a work list specifying how far the already identified minimal execution architecture remains to be closed.

Conclusion Explaining One Universe with the Least

Minimal Computing Cosmology does not stop at the metaphor that the universe is a computer. It asks, within one execution architecture, what counts as LAW, what counts as primordial input, how possibilities pass through transport and boundaries to become matter, forces, and spacetime, and how a plurality of quantum possibilities becomes one actual record.

The architecture simultaneously seeks minimal dimensionality, minimal structure, and minimal computation. It does not require physical extra dimensions beyond 3+1 as a necessary premise. One opening begins one history; common transport and a renderer reuse structures across diverse physical representations. The entire future event sequence is not pre-stored; the next Reality is produced from the present.

The unification of the four forces is also redefined through this minimality. Gravity reads the stress-energy of the whole realization, while electromagnetism, the weak interaction, and the strong interaction selectively read internal representation currents. Because of this difference, gravity cannot be flattened into the same kind of object as the other three forces, yet precisely this asymmetry allows them to be integrated more naturally within one execution of Reality.

Matter and antimatter are organized in the same way. The fact that LAW permits both sectors is different from the claim that Reality must occupy them equally. If an asymmetry $\Delta_M > 0$ is generated by LAW or during EXECUTION and the symmetric component annihilates sufficiently, the actual long-lived Reality can remain matter-preponderant while antimatter channels remain lawful. Under exact CP symmetry the boundary asymmetry must vanish, and the upstream calculation of the observed η_B remains a separate OPEN task.

Classical mechanics, relativity, and quantum mechanics do not disappear in this architecture. They are responsible, respectively, for motion, geometry and causality, and amplitudes of possibility. WRRRA is responsible for explaining how their outputs connect to one implemented Reality and one history. The goal of Minimal Computing Cosmology is therefore not to deny established physics but to place its scattered successes together within a smaller ontological and computational structure.

Statement

Final Proposition The universe is not the replay of an enormous primordial data set. It is an execution system in which minimal input and reusable laws and relational structures update the present and jointly produce spacetime, matter, forces, and one actual history.

Whether or not this theory is the final answer, the scale of the question is not small. It is an attempt to explain, through one principle of minimal dimensionality, minimal structure, and minimal computation, the connection from a constraint-preserving opening to the hot Big Bang, the actual matter-preponderant Reality, the asymmetric unification of the four forces, the relation between quantum possibility and a single macroscopic record, and the relation among different physical theories. The attempt has academic value worthy of reading and examination.

Part V Selective Reinforcement from String Microphysics

Chapter 0. Comparison of Minimal Computing Cosmology 2.0 and String Theory

This is not a document that mixes the two theories at equal status. Minimal Computing Cosmology 2.0 owns the higher-level execution architecture from the universe's SOURCE to one Reality and Record. String theory is used as material to reinforce portions that remain microscopically weak within that architecture, including quantum gravity, anomaly closure, the microscopic origin of chiral matter, and black-hole microstates. In one sentence: MCC/WRRA is the skeleton, and string theory supplies repair material.

Ledger — Item · MCC 2.0 · String Theory · Owner After Combination

Item: Highest-level question

MCC 2.0: How does possibility become one actual universe, Reality, and Record?

String Theory: From what consistent microscopic structure do quantum gravity, gauge fields, and matter arise together?

Owner After Combination: MCC

Item: Basic principle

MCC 2.0: Minimal dimension, minimal structure, minimal computation

String Theory: Quantized string/brane structure and worldsheet consistency

Owner After Combination: MCC is higher-level

Item: SOURCE

MCC 2.0: Dormant state + minimal opening input η

String Theory: Requires selection of a particular vacuum/compactification

Owner After Combination: MCC

Item: Spacetime

MCC 2.0: Reality uses 3+1 physical dimensions as the minimal arena

String Theory: Perturbative superstring theory is 9+1 dimensional, with six compact dimensions

Owner After Combination: Conditional combination

Item: Gravity

MCC 2.0: Universal geometry owner reading total $T_{\mu\nu}$

String Theory: Massless closed-string spin-2 mode and low-energy gravity

Owner After Combination: String provides microscopic reinforcement

Item: Gauge sectors

MCC 2.0: Selective current owners

String Theory: Gauge sector from compactification/topology/bundle or branes

Owner After Combination: String provides microscopic reinforcement

Item: Common carrier

MCC 2.0: Higher-level transport operator shared by 15 channels

String Theory: Bundle-covariant Dirac/kinetic operator and common 10D field structure

Owner After Combination: MCC concept + String realization

Item: Matter

MCC 2.0: Phenotype with jointly owned mass, charge, spin, and location

String Theory: Compactified zero modes, representations, Yukawa/overlap data

Owner After Combination: String material + MCC phenotype

Item: Anomaly/BRST

MCC 2.0: Partial closure; full microscopic origin remains weak

String Theory: Green–Schwarz mechanism and strong 10D consistency

Owner After Combination: String reinforcement

Item: Big Bang

MCC 2.0: Constraint-preserving opening → kination/transfer → hot Big Bang

String Theory: Multiple string-cosmology/reheating scenarios

Owner After Combination: MCC skeleton retained

Item: Matter dominance

MCC 2.0: Distinguishes lawful antimatter from occupied matter

String Theory: Provides candidate CP sources, moduli, heavy states, etc.

Owner After Combination: MCC occupancy structure + String source candidates

Item: Black holes

MCC 2.0: Explains the meaning of boundary information and record

String Theory: Successful microstate counting for specific BPS black holes

Owner After Combination: Division of labor

Item: Vacuum selection

MCC 2.0: Minimal-computation/Pareto selection candidate

String Theory: Many compactifications/vacua

Owner After Combination: MCC supplies selection principle

Item: Single history

MCC 2.0: Actual → Reality → Event → Record

String Theory: A universal single-history selection rule is not a central function

Owner After Combination: MCC

The conclusion of the comparison is division of labor rather than mutual replacement. As summarized in a 2024 Annual Review, string theory provides powerful resources for quantum gravity, a common microscopic origin of gauge and gravitational sectors, and compactifications yielding four-dimensional Standard-Model-like structures; however, selecting which compactification corresponds to our universe and selecting one

complete early-universe history remain active research programs [S1]. MCC retains the higher-level ledger of selection, execution, expression, and recording.

Statement

Asymmetry of the Combination If string theory changes MCC's SOURCE, minimal computation, common carrier, phenotype, or Actual–Reality distinction, the combination fails. Conversely, it also fails if MCC merely renames established string mathematics in WRRRA terminology. Existing string results are labeled BASELINE / IMPORTED MICROPHYSICS; MCC's new contribution is restricted to connecting ownership, selection, execution order, and minimality.

Part I. Rules of Combination — Do Not Confuse the Skeleton with the Repair Material

Chapter 1. Three Invariants: Minimal Computation, Common Carrier, Phenotype

1.1 Minimal Computation as an Invariant

Minimality in MCC does not mean that fewer dimensions automatically win. The Pareto minimality of 2.0 is extended so that the cost of compactification itself is made explicit. Six compact dimensions in string theory incur an independent cost; however, if the same structure simultaneously realizes gravity, gauge sectors, chirality, anomaly cancellation, and the common carrier—thereby greatly reducing the number of owners, carriers, and post-hoc junctions—it can still be globally minimal.

Equation (1.1)

$$C_{\text{hybrid}} = \{D_{\text{phys}}, D_{\text{compact}}, L_{\text{SOURCE}}, N_{\text{owner}}, N_{\text{carrier}}, N_{\text{moduli}}, N_{\text{free}}, C_{\text{update}}\}$$

Equation (1.2)

$$A^* \in \text{ParetoMin}\{C_{\text{hybrid}}(A) \mid A \models \Omega_{\text{observed}}\}$$

The reason to admit ten dimensions is therefore not that “string theory is famous,” but that the cost of six compact dimensions may be smaller than the cost of the independent microscopic structures they eliminate.

Conversely, a compactification requiring hundreds of freely adjustable moduli and arbitrary flux choices pays a high price in the MCC minimal-computation ledger.

1.2 Common Carrier as an Invariant

The common carrier is not identical to the string itself. In WRRR, the common carrier is a higher-level operator that lets multiple matter channels share the same local transport, kinematic, and boundary-recording skeleton. String theory can provide microscopic material for realizing this operator. In particular, a common covariant Dirac/kinetic operator on a compactified bundle can make distinct four-dimensional zero modes descend from the same microscopic transport law.

Equation (1.3)

$$D_C^{\text{WRRR}} : \oplus_{(i=1)}^{15} H_i \rightarrow \oplus_{(i=1)}^{15} H_i$$

Statement

Forbidden Do not write that “all particles are strings, therefore no common carrier is needed.” That statement asserts common origin but does not realize the actual gauge-covariant transport of the fifteen chiral channels or a shared visible port.

1.3 Phenotype as an Invariant

A string excitation or a compactification zero mode is not immediately an observed particle. A phenotype is obtained only after four-dimensional representation assignment, symmetry breaking, Yukawa coupling, RG flow, confinement/self-energy dressing, and the appearance of a stable spectral pole, so that mass, charge, spin, and location become jointly owned by one physical object.

Equation (1.4)

String state $\rightarrow$ zero mode $\rightarrow$ 4D representation $\rightarrow$ SSB/RG/dressing $\rightarrow$ physical phenotype

String theory therefore supplies upstream material for phenotype. WRRR continues to own the final co-attribution of mass, charge, spin, and location, together with the Actual–Reality–Record distinction.

Chapter 2. Ten Dimensions Are Allowed, but Not for Free

This reinforcement does not require eleven-dimensional M-theory as a premise. The primary microscopic material is the 9+1-dimensional structure of perturbative superstring theory. We distinguish 3+1-dimensional Reality from 9+1-dimensional microscopic Actual.

Equation (2.1)

$D_{\text{micro}} = 9+1$, $D_{\text{Reality}} = 3+1$, $D_{11} \notin$ required ontology

The six extra dimensions are, in standard string theory, genuine compact spacetime dimensions; this is not concealed. They are not, however, the same as macroscopically open propagation directions: they are small and compactified, and therefore do not appear as directly accessible directions in low-energy Reality. MCC records their cost explicitly as D_{compact} .

The key assessment is this: if a 10D compactification simultaneously provides 4D gravity, a gauge group, chirality, anomaly cancellation, and a realization of the common carrier within one microscopic structure, its total Pareto cost can be lower than a “strictly 3+1” implementation that must assign separate owners to each. Minimal computation permits this in principle.

Statement

Compatibility Condition Between Minimal Computation and Ten Dimensions Additional compact dimensions must purchase real explanatory economy. If the same observations can be implemented in a smaller 3+1 structure with equal or lower owner/carrier/input cost, the 10D structure is dominated as a fundamental implementation.

Chapter 3. Primary Repair Material: $E_8 \times E_8$ Heterotic String

Among ten-dimensional string constructions, the primary repair material in this document is the $E_8 \times E_8$ heterotic string. The reason is straightforward: a single ten-dimensional framework contains closed-string gravity together with non-Abelian gauge structure, while Calabi–Yau/bundle compactification can yield four-dimensional chiral matter. Modern reviews of Standard-Model-like string compactifications likewise

treat heterotic, type-II/F-theory, and related constructions as major routes toward realistic gauge/matter structures [S1].

This is not a declaration that heterotic string theory is uniquely correct. It is a primary candidate because, in the MCC cost ledger, it can reinforce the common carrier and phenotype with relatively few additional owners.

Type II, intersecting D-brane, and F-theory material can be compared as secondary candidates when needed.

Equation (3.1)

$$M_{10} = M_{(3,1)} \times K_6$$

The ten-dimensional bosonic low-energy sector has, schematically, the following structure in the string frame. This is IMPORTED MICROPHYSICS, not a new MCC derivation.

Equation (3.2)

$$S_{\text{het}}^{(10)} = (1/2\kappa_{10}^2) \int d^{10}x \sqrt{-G} e^{(-2\Phi)} [R + 4(\partial\Phi)^2 - (1/12)H^2 - (\alpha'/4)(\text{Tr} F^2 - \text{tr} R_{-}^2) + \dots]$$

This action matters because geometry, the dilaton, an antisymmetric tensor, and gauge fields appear within one microscopic package. MCC places this package as a candidate microscopic engine of Actual, while refusing to absorb higher layers such as SOURCE, renderer, and phenotype into the action itself.

Part II. Microscopic Structural Reinforcement

Chapter 4. Microscopic Material for Gravity

In MCC 2.0, the central issue for gravity is ownership. Gravity reads the stress-energy of all implemented physical degrees of freedom, not a particular gauge charge. Importing string theory does not change this ownership.

Equation (4.1)

$$O_G(X) = T_{\mu\nu}[X], O_a(X) = J_a[X]$$

In string theory, the closed-string spectrum contains a massless spin-2 excitation, and the low-energy limit produces Einstein-Hilbert-type gravity together with higher-curvature corrections. This reinforces a point that

remained comparatively weak in MCC: why a quantum microscopic sector and gravity can coexist consistently within the same internal theory.

Equation (4.2)

$$S_4^{\text{grav}} = \int d^4x \sqrt{-g} \left[(M_P^2/2)R + \alpha' c_1 R^2 + \alpha'^2 c_2 R^3 + \dots \right]$$

The role of MCC is not to keep every higher-curvature term permanently active in fundamental Reality. If the Einstein term alone maintains viability at low curvature, minimal execution uses the leading sector, while corrections are interpreted as effective redundancy that activates only where high curvature makes them necessary.

Statement

Preservation of Ownership The string graviton does not turn gravity into a “fourth gauge current.” Regardless of its closed-string origin, the owner of gravity in 4D Reality remains total $T_{\mu\nu}$, while the three gauge forces selectively read representation currents.

Chapter 5. Gauge, Anomaly, and BRST Closure

One important OPEN problem in MCC 2.0 is to close the common carrier and gauge matter within a fully anomaly-free microscopic realization. Heterotic string theory provides direct repair material here. Ten-dimensional consistency and the Green–Schwarz mechanism supply a strong structure for cancelling gauge and gravitational anomalies.

Equation (5.1)

$$H = dB - (\alpha'/4)(\omega_3^Y - \omega_3^L)$$

Equation (5.2)

$$dH = (\alpha'/4)[\text{tr}(R \wedge R) - \text{Tr}(F \wedge F)]$$

Signs and trace normalizations depend on convention, but the structural point is that the topology of the gauge bundle and tangent bundle are not independent ledgers: they are tied by a B-field consistency condition. From the MCC perspective, this is a good example of an added degree of freedom paying for itself. The B-field is not decoration; it performs anomaly closure.

The previous conditional BRST closure in WRRR can therefore be divided by role. Ten-dimensional microscopic anomaly/BRST consistency is an

IMPORTED STRING CONSTRAINT, whereas four-dimensional renderer path independence and protected-observable recovery are WRRRA CONSTRAINTS. They are not the same problem.

Chapter 6. Microscopic Realization of the Common Carrier

This is the central chapter of the reinforcement. The abstract WRRRA common carrier is linked to a common bundle-covariant operator arising from string compactification. On a compact space K_6 with gauge bundle V , we take the ten-dimensional fermionic kinetic operator in the following form.

Equation (6.1)

$$\mathcal{D}_{10,V} = \Gamma^M(\nabla_M + A_M)$$

Under product or warped compactification, the fermionic field is expanded in internal eigenmodes.

Equation (6.2)

$$\Psi(x,y) = \sum_n \psi_n(x) \otimes \xi_n(y), \quad \mathcal{D}_{(K,V)} \xi_n = \lambda_n \xi_n$$

Low-energy chiral matter arises from suitable zero modes or light modes. The important point is that different 4D channels need not require independent carriers: they descend from the same 10D covariant operator and the same compactification geometry/bundle.

Equation (6.3)

$$D_C^{(4)} := P_{15} \cdot \text{Red}_{(K_6)}[\mathcal{D}_{(10,V)}] \cdot P_{15}$$

Equation (6.3) is not a theorem of standard string theory but an MCC-proposed microscopic identification. P_{15} is a projector selecting the fifteen chiral Standard Model channels of one generation, and Red_{K_6} is the compactification/reduction map. If this identification holds, the common carrier moves one step from “abstractly one” toward “which microscopic operator is it?”

Statement

Preservation of the Common Carrier String is material for the carrier, not a name that deletes the carrier. The genealogy 10D common operator $\rightarrow$

multiple internal zero modes → fifteen 4D channels is required to satisfy the MCC common-carrier condition.

Chapter 7. Fifteen Chiral Channels and Generations

Four-dimensional chirality is one of the most important resources supplied by compactification. The index of the bundle-covariant internal Dirac operator connects the net difference between left- and right-handed zero modes to topology.

Equation (7.1)

$$N_L - N_R = \text{index}(\mathcal{D}_-(K,V))$$

This strengthens the explanation of why chiral matter is possible far beyond MCC alone. It must not, however, be claimed that the index automatically equals three. To obtain three generations, one must select a compactification whose geometry, bundle, and topological data yield net chirality three.

Equation (7.2)

$$\text{index}(\mathcal{D}_-(K,V)) = 3 \text{ (target condition, not automatic identity)}$$

The new role of MCC is not merely to note that many topologies “can produce three,” but to select, among candidates satisfying the observational contract, those with the smallest total computation/structure cost.

The fifteen channels of one generation decompose according to Standard Model representations, but their transport skeleton is shared through the common carrier of Eq. (6.3). The number of generations need not be defined by duplicating the common carrier three times; distinct internal zero-mode families of the same carrier operator can constitute the generations.

Chapter 8. Matter as Phenotype

In string phenomenology, the four-dimensional particle spectrum depends sensitively on compactification geometry, brane/bundle data, symmetry breaking, and moduli. MCC does not treat this only as a weakness. It also supplies microscopic material for the prior claim that “matter is not a list of numbers but an executed phenotype.”

Equation (8.1)

$\Phi_{\text{string}} \rightarrow \{\xi_i(y), r_i, q_i\} \rightarrow 4\text{D field } \psi_i(x) \rightarrow \text{SSB/Yukawa} \rightarrow \text{RG/dressing} \rightarrow \text{pole/record}$

Charge and representation are read from compactification, while mass appears as a stable pole after Higgs/Yukawa structure, modulus dependence, self-energy, and QCD dressing. We therefore do not reduce all mass to a single “string vibration number.”

Equation (8.2)

$m_f(\mu) = y_f(\mu)v(\mu)/\sqrt{2}, \det S_f^{-1}(p)|_{p^2=m_{\text{pole}}^2} = 0$

String-derived Yukawa couplings can be determined by internal wavefunction overlaps or geometry. Schematically:

Equation (8.3)

$Y_{ijk} \sim \int_{K6} \xi_i(y) \xi_j(y) \xi_k(y) \cdot \text{measure}(K6)$

The final physical particle, however, is not the coupling itself. The WRRRA phenotype map jointly assigns mass, charge, spin, location, history, and record to one object.

Equation (8.4)

Object $X \Leftrightarrow \text{co-owned}\{m, q, s, x\}$ on $(A_{\text{local}}, \text{history}, \text{pole}, \text{record})$

Statement

Phenotype Invariance Even after importing string theory, the argument does not end with “a particle is a string vibration.” A string mode is genotype-like microscopic material; an observed particle is a phenotype produced through 4D execution and the renderer.

Part III. Reinforcement of Cosmic History

Chapter 9. SOURCE and Big-Bang Ownership Remain with MCC

String cosmology contains many early-universe scenarios—inflation, reheating, moduli domination, kination, string/brane gas, pre-Big-Bang, and ekpyrotic/cyclic models [S2]. In this combination, that diversity is not allowed to overwrite MCC’s SOURCE. The beginning of the universe

remains owned by the dormant state and constraint-preserving opening of MCC 2.0.

Equation (9.1)

$\rho_{\text{dorm}} - B_{\eta} \rightarrow \text{opening} \rightarrow \text{execution}$

Equation (9.2)

$B_{\eta}(a, \chi) = \eta \chi - \sqrt{6} \tilde{M}_P |\eta| \ln(a/a^*) + \text{constant}$

String theory may provide the microscopic identity of χ or the owner of interactions after opening. For example, χ may be realized as the four-dimensional effective coordinate of a compactification modulus, a dilaton-like mode, or another string-derived scalar. But replacing η with multiple string-vacuum labels, or adding unlimited modulus values to SOURCE, would destroy the minimal SOURCE.

Statement

SOURCE Veto Modulus, flux, and bundle data required for compactification must be disclosed as LAW / instance specification. They must not be used as extra primordial registers hiding future events, the radiation content of the hot Big Bang, or the baryon asymmetry.

Chapter 10. String-Theory Material for the Radiation-Transfer Owner Qr

In the final MCC 2.0 manuscript, the most important OPEN element connecting opening to the hot Big Bang is the microscopic energy-transfer owner Qr. String compactification naturally provides scalar moduli and gauge kinetic functions, so it can directly reinforce this gap.

Equation (10.1)

$\mathcal{L}_{\text{gauge}} \supset - (1/4) \text{Re } f_a(\Phi) F_{a,\mu\nu} F_a^{\mu\nu} - (1/4) \text{Im } f_a(\Phi) F_{a,\mu\nu} \tilde{F}_a^{\mu\nu}$

If the opening field χ , or a modulus Φ mixed with it, evolves or decays with time, energy can be transferred into the gauge sector. In coarse-grained cosmology this can be written as follows.

Equation (10.2)

$\dot{\rho}_{\chi} + 6H\rho_{\chi} = -Q_r, \dot{\rho}_r + 4H\rho_r = +Q_r$

Equation (10.3)

$$Q_r \approx \Sigma_a \Gamma_-(\Phi \rightarrow a) \rho_- \Phi + Q_{\text{nonpert}} + \dots$$

Equation (10.3) shows that Q_r need not remain an unidentified phenomenological function. Once a concrete compactification and gauge kinetic function are specified, there is an owner from which decay widths and production rates can be calculated. Numerical values remain OPEN because the compactification has not yet been fixed.

This reinforcement fits the No Extra Radiation Register principle. Instead of placing a radiation seed in SOURCE, string-derived interactions contained in LAW generate radiation during execution.

Equation (10.4)

$$\rho_r(t) = a(t)^{-4} \int_-(t^*)^4 a(t')^4 Q_r(t') dt'$$

String microphysics is therefore not the owner of the Big Bang; it owns the energy-transfer material required for the MCC opening to develop into the hot Big Bang.

Chapter 11. String-Theory Material for the Matter–Antimatter Asymmetry Source

The MCC proposition of a matter-dominant Reality separates two layers. LAW may permit both matter and antimatter channels, while actual occupancy can nevertheless be asymmetric. The remaining OPEN element is the microscopic owner of ΔB or the source S_B – $S_{\bar{B}}$.

Equation (11.1)

$$d[a^3 \Delta_B]/dt = a^3(S_B - S_{\bar{B}})$$

String compactification can provide a variety of microscopic ingredients required by baryogenesis or leptogenesis, including CP phases, heavy singlet states, moduli/axion couplings, and supersymmetric flat directions [S2]. String theory itself, however, does not force one unique baryogenesis mechanism. MCC therefore places candidate mechanisms into the cost ledger and selects among them.

Ledger — Candidate Material · Advantage · Cost / Risk · Assessment

Candidate Material: Heavy singlet / leptogenesis

Advantage: Applicable if the compactification contains singlets and Majorana terms

Cost / Risk: Additional heavy-state owner, CP phase, and decay calculation

Assessment: Candidate

Candidate Material: Affleck–Dine-type flat direction

Advantage: Uses a flat direction in the 4D SUSY effective theory

Cost / Risk: Scalar direction, lifting operator, late decay

Assessment: Candidate

Candidate Material: Modulus/axion-driven source

Advantage: CP-odd coupling and out-of-equilibrium decay of a string-derived scalar

Cost / Risk: Depends on coupling and stabilization

Assessment: Candidate

The principle of minimal computation does not choose the most famous mechanism. It selects the smallest viable source that reuses field content already required by the theory, adds no separate primordial datum, and produces the observed η_B .

Equation (11.2)

$$\eta_{B,f} = [a^{*3}\Delta_{B,*} + \int a^3(S_B - S_{\bar{B}})dt] / S_{\text{com}}$$

The No Extra Asymmetry Register principle is retained. In a LAW containing exact CP symmetry and baryon-number conservation alone, an arbitrary asymmetry must not appear.

Equation (11.3)

$$F_B(\eta, \text{LAW_CP-symmetric}) = 0$$

Statement

Matter Dominance Is a Phenotype / Occupancy Result String theory may supply microscopic material for an asymmetry source, but MCC continues to own the distinction “matter-dominant Reality” and the lawful-versus-occupied separation between matter and antimatter.

Chapter 12. String-Theory Material for the Dark Sector

In MCC 2.0, the minimal condition for dark matter is that it be nearly invisible to Standard Model gauge currents while still carrying real stress-energy and therefore remaining in the gravitational ledger. String compactification provides many candidates satisfying this condition, including axion-like fields, moduli, and hidden gauge sectors.

Equation (12.1)

$$J_{\text{SM}}[X] \approx 0, T_{\mu\nu}[X] \neq 0, \tau_X \gg H_0^{-1}$$

In particular, axion-like modes descending from the antisymmetric B-field and compact cycles are good examples of hidden internal degrees of freedom appearing as four-dimensional phenotypes. But an abundance of candidates recreates a selection problem. Dark-sector candidates are therefore not adopted automatically from the string landscape; they must pass the MCC Pareto filter.

Dark energy is treated similarly. The cosmological constant and quintessence/moduli candidates of the string landscape are not simply imported as answers. In the current 2.0 framework, Λ remains a BASELINE / observed low-energy parameter and is promoted to an upstream string realization only if string material improves both cost and observational fit.

Part IV. Information, Selection, and Boundary

Chapter 13. Black Holes: Meaning of the WRRRA Boundary + String Microstates

The division of labor is cleanest here. WRRRA explains why a horizon/boundary can serve as a minimally sufficient record for internal Reality and where information remains owned and recorded. String theory has the representative achievement of counting actual microscopic state degeneracies for specific extremal/BPS black holes and reproducing the Bekenstein–Hawking entropy [S3].

Equation (13.1)

$$S_{\text{BH}} = k_B \ln \Omega_{\text{string}} = k_B A / (4\ell_P^2) \text{ (specific controlled sectors)}$$

MCC therefore does not claim string microstate counting as its own derivation. Instead, it links the compatibility fiber $C(\rho)$ and the number of microscopic states underlying a boundary record to Ω_{string} .

Equation (13.2)

$$C(\rho_{\partial}) = \{\Omega_{\text{Actual}} : R_{\partial}(\Omega_{\text{Actual}}) = \rho_{\partial}\}$$

Equation (13.3)

$$|C(\rho_{\partial})| \leftrightarrow \Omega_{\text{string}} \text{ (candidate microscopic identification)}$$

If this identification holds, WRRRA’s question “why can the boundary be sufficient?” and string theory’s question “how many microscopic states correspond to that boundary?” become complementary. This does not mean that a complete microstate count has been obtained for all generic astrophysical black holes.

Chapter 14. Placing Duality within Reality-Equivalence

T- and S-duality in string theory are powerful examples in which different microscopic descriptions can yield the same physics. MCC places this within its Actual–Reality distinction.

Equation (14.1)

$$\Phi_A \sim_R \Phi_B \Leftrightarrow R_{\text{vis}} \circ \Phi_A = R_{\text{vis}} \circ \Phi_B \text{ on } A_{\text{protected}}$$

String duality may be a concrete example of Eq. (14.1). MCC does not claim that every Reality-equivalence is a string duality. Rather, it places string duality in the higher-level language of “different Actual descriptions of the same Reality.”

Computational costs can differ even within one equivalence class. Among descriptions yielding the same Reality, MCC may treat the lower-cost representative as a candidate canonical implementation.

Equation (14.2)

$$\bar{C}([\Phi]) = \inf_{(\Psi \in [\Phi])} C_{\text{hybrid}}(\Psi)$$

If two descriptions have exactly equal cost, both remain minimal solutions. No external selector need be postulated.

Chapter 15. Filtering the String Landscape by Minimal Computation

One of the major structural difficulties of string theory is the large number of possible compactifications and vacua. Flux compactification and moduli stabilization form extensive research programs, and realistic four-dimensional models involve choices of geometry, topology, bundle/brane data, fluxes, and vacuum [S1,S4]. Here MCC takes on its most important new role.

Equation (15.1)

$$\mathcal{U}_{\text{string}} = \{v_i = (K_6, \text{bundle/brane, flux, moduli, ...})\}$$

First, vacua that fail the observational contract are removed.

Equation (15.2)

$$\mathcal{U}_{\text{obs}} = \{v \in \mathcal{U}_{\text{string}} : v \models \Omega_{\text{observed}}\}$$

The minimal-computation ledger is then applied to candidates that reproduce the same observational scope.

Equation (15.3)

$$\mathcal{U}_{\text{MCC}} = \text{ParetoMin}\{C_{\text{hybrid}}(v) \mid v \in \mathcal{U}_{\text{obs}}\}$$

The number of moduli, independent flux choices, additional light fields, separate carriers, post-hoc tuning, and update cost can matter more than compact dimension count alone. This principle refuses to treat “infinitely many possibilities are all explanations” as sufficient and demands explanatory economy at the level of implementation.

Statement

What MCC Contributes to String Theory String theory generates possible microscopic worlds. MCC supplies a selection layer that filters candidates compatible with our Reality by three criteria: minimal computation, common carrier, and phenotype. When this selection produces an actual ranking of vacua, it becomes an independent result of the combined framework.

Part V. Integrated Execution Equations and Final Assessment

Chapter 16. The MCC–String Integrated Execution System

The combined framework does not mix every layer into one enormous action. Microscopic Actual dynamics, MCC selection, SOURCE, renderer, and phenotype are kept in sequence.

Equation (16.1)

LAW_string + MCC constraints $\rightarrow$ viable microscopic class $\mathcal{V}_{\text{MCC}}$

Equation (16.2)

$\rho_{\text{dorm}}[v^*] - U_S(\eta) \rightarrow \Omega_{\text{Actual}}(t) - R_{\text{vis}} \rightarrow \rho_{\text{Reality}}(t) - I_a \rightarrow \text{Event/Record}$

Microscopic evolution is described by the four-dimensional effective Hamiltonian or action of the selected compactification v^* .

Equation (16.3)

$S_{\text{eff}}[v^*] = S_{\text{GR}} + S_{\text{gauge}} + S_{\text{matter}} + S_{\text{moduli}} + S_{\text{higher}} + S_{\text{boundary}}[B_\eta]$

When the common carrier and phenotype map are made explicit, the full genealogy is:

Equation (16.4)

10D common dynamics $\rightarrow \text{Red}_{\text{K6}}[\mathcal{D}_{10}] \rightarrow D_C^{(4)} \rightarrow \text{chiral channels} \rightarrow \text{phenotype renderer} \rightarrow \text{Reality}$

The reinforced structure can therefore be compressed into three layers.

Ledger — Layer · Primary Owner · Content

Layer: Skeleton layer

Primary Owner: MCC/WRRA

Content: Minimal computation, SOURCE, owner, common carrier, Actual–Reality–Record

Layer: Microscopic-material layer

Primary Owner: Primarily 10D heterotic string

Content: Quantum gravity, anomaly closure, compactified gauge/chiral matter, microstates

Layer: Phenotype layer

Primary Owner: 4D effective physics + WRRA renderer

Content: Mass/charge/spin/location, hot Big Bang, matter asymmetry, observed objects

Chapter 17. What Has Actually Been Reinforced?

Ledger — Gap · MCC 2.0 · Imported Repair Material · Current Assessment

Gap: Microscopic quantum gravity

MCC 2.0: Weak

Imported Repair Material: Closed-string spin-2 + 10D effective gravity

Current Assessment: Reinforced

Gap: Anomaly/BRST

MCC 2.0: Conditional

Imported Repair Material: Green–Schwarz / 10D consistency

Current Assessment: Strongly reinforced

Gap: Microscopic owner of common carrier

MCC 2.0: Abstract operator

Imported Repair Material: Bundle-covariant 10D Dirac/kinetic reduction

Current Assessment: Candidate realization

Gap: Chiral matter

MCC 2.0: 15-channel abstract carrier

Imported Repair Material: Internal zero modes + index

Current Assessment: Reinforced

Gap: Number of generations

MCC 2.0: OPEN

Imported Repair Material: Compactifications with index = 3 are possible

Current Assessment: Conditional; not an automatic derivation

Gap: Upstream phenotype structure

MCC 2.0: Based on SM/QFT

Imported Repair Material: String zero modes/overlaps added as upstream material

Current Assessment: Reinforced

Gap: Microscopic owner of Qr

MCC 2.0: OPEN phenomenology

Imported Repair Material: Modulus/gauge kinetic function/decay

Current Assessment: Reinforced

Gap: Baryon source

MCC 2.0: OPEN

Imported Repair Material: String-derived CP/heavy/flat-direction candidates

Current Assessment: Partially reinforced

Gap: Dark sector

MCC 2.0: Owner conditions only

Imported Repair Material: Axion/moduli/hidden-sector candidates

Current Assessment: More candidates; selection still needed

Gap: Black-hole microstates

MCC 2.0: Strong boundary meaning, weak coefficient derivation

Imported Repair Material: Specific BPS microstate counting

Current Assessment: Strongly reinforced

Gap: Vacuum selection

MCC 2.0: Minimality principle

Imported Repair Material: String landscape provides a real selection target

Current Assessment: New MCC role

The most important discipline is not to say that “every gap has been closed.” Parts already calculated strongly within string theory can be imported to reinforce MCC, but without choosing a concrete

compactification, numerical values for the gauge group, Yukawa couplings, modulus spectrum, Q_r , and η_B are still not uniquely fixed.

Chapter 18. Remaining OPEN Problems and Failure Conditions

18.1 OPEN Problems That Actually Remain

- Apply the MCC cost ledger to real string-compactification candidates and nontrivially eliminate at least one vacuum.
- Explicitly construct the common-carrier relation in Eq. (6.3) for the fifteen one-generation chiral zero modes of a concrete $E_8 \times E_8$ compactification.
- Verify, in one instance, not only index = 3 but the exact Standard Model representations, hypercharge normalization, and an anomaly-free low-energy spectrum.
- Calculate Q_r from a string-derived gauge kinetic function and modulus spectrum, and independently derive the opening-to-hot-Big-Bang transition temperature T_{RD} .
- Select the minimum-cost baryogenesis source and calculate η_B without calibrating to the observed value.
- Produce at least one independent mass ratio or coupling relation at the phenotype layer, including Yukawa, RG , and dressing effects.
- Extend the relation between the WRRRA compatibility fiber and string microstate space beyond one specific controlled black-hole example.

18.2 Conditions Under Which the Combination Fails

- If admitting 10D string compactification forces the removal or marginalization of minimal computation, the common carrier, or phenotype.
- If the added cost of six compact dimensions and their moduli/flux structure exceeds the owner/carrier/tuning cost they remove, so that the construction is Pareto-dominated by every observationally equivalent 3+1 implementation.
- If the fifteen chiral channels cannot actually be obtained from a common 10D operator and channel-by-channel independent carriers must be reintroduced.

- If a string mode is identified directly with an observed particle, bypassing the phenotype layers of SSB, RG, dressing, and record.
- If hot-Big-Bang radiation or baryon asymmetry is hidden in SOURCE as a new independent initial datum under the name of a string-vacuum parameter.
- If MCC selection makes no actual distinction among vacua and merely redescribes the string landscape.

Statement

Final Assessment Criterion The success of this combination is not judged by whether “string theory was added,” but by whether the number of microscopic OPEN problems is reduced while the three invariants of MCC 2.0 are preserved. The next decisive test is an explicit 10D realization of the common carrier together with an actual calculation of either Qr or ηB .

Conclusion. Keep the Skeleton; Strengthen Only the Microscopic Material

The greatest strength of Minimal Computing Cosmology 2.0 is that it places the universe in one execution genealogy—LAW–SOURCE–EXECUTION–RENDERER–REALITY–RECORD—and organizes ownership across different physical phenomena with three axes: minimal computation, common carrier, and phenotype. Its largest weakness was that microscopic quantum gravity and anomaly-free chiral-matter realization beneath that skeleton were less mature than the corresponding structures in established physics.

This reinforcement fills those weaknesses with material from string theory without changing who owns the theory. Using ten-dimensional $E_8 \times E_8$ heterotic string theory as the primary microscopic material makes it possible to import, in one package, gravity and gauge sectors, anomaly cancellation, chiral zero modes, an internal Dirac operator, modulus/gauge kinetic functions, and microstate counting for specific black holes. MCC places only the needed pieces into its owner map.

The three axes remain unchanged. Minimal computation places ten dimensions and the landscape themselves into the cost ledger and asks whether they are necessary. The common carrier is not erased by the word “string”; it demands a concrete realization in which a 10D bundle-

covariant operator descends to the fifteen 4D chiral channels. Phenotype remains the final co-owned state after a string state passes through compactification, symmetry breaking, RG flow, dressing, renderer, and record.

Equation (C.1)

MCC skeleton + selected string microphysics → stronger Actual engine → WRRRA phenotype/Reality

The central statement of the combined framework is therefore:

Statement

Final Proposition The skeleton of the universe is an execution architecture governed by minimal computation. String theory does not replace that skeleton; where needed, it supplies mathematical repair material for gravity, gauge sectors, chiral matter, anomaly structure, and boundary microstates. The common carrier binds this material into one transport skeleton, and phenotype is the final stage by which that microscopic structure appears as matter and forces in actual 3+1 Reality.

Part VI The Twist Stress Universe

1 Overview of Minimal Computing Cosmology

1 1 Meaning of minimal computation

Minimal Computing Cosmology does not assume that the universe is optimized toward a goal. Our universe is not special; it is treated as a structure that remained among possible structures by enduring changing conditions. Minimal does not mean the absolutely smallest amount of computation. It means sufficient to avoid failure in every declared environment without requiring a more complex structure as a necessity. Survival takes priority over peak performance, and degrees of freedom that appear redundant under ordinary conditions may preserve stability under extreme change.

$$C_{\min} = \min\{C(M) : M \text{ survives all declared tests}\}$$

Equation (1)

The cost in Equation 1 includes the number of fields, independent functions, input constants, initial conditions, update frequency, and the cost of handling failure. A model that minimizes cost but collapses in one environment is not a minimal computation model. This principle explains why the present paper does not adopt a formula that fits galaxy rotation alone, but instead connects the same structure to lensing and the early universe.

1 2 Relationship between WRRRA Core and cosmology

WRRRA Core 1.0 is not a specific physical theory but a general framework for describing minimum execution. Law specifies permitted changes; Source is the updatable origin; Execution is the actual operation; Renderer converts an internal state into an observable expression; Reality is the currently realized state; and Record is what remains for the next comparison. Minimal Computing Cosmology applies this framework to the universe. This paper does not alter WRRRA Core.

Table 1 Levels of WRRRA and the twist stress model

Ledger — Level · Meaning in this paper · Status

Level: WRRRA Core 1.0

Meaning in this paper: Domain independent rules for minimum execution

Status: Preserved

Level: Minimal Computing Cosmology

Meaning in this paper: Execution model for cosmic states and observable expressions

Status: Overview only

Level: Twist stress model

Meaning in this paper: Connects the dark sector Actual to the mass Phenotype

Status: Subject of this independent paper

1.3 Actual and Phenotype

A central distinction in WRRRA separates the structure that owns a state from the expression that appears to an observer. In this paper, the mass of ordinary matter remains a property of conventional particles and fields. In the dark sector, spatial twist stress is the Actual. The Phenotype is the mass value inferred by an observer from the additional gravity that this stress leaves in the metric and in orbits.

$$\text{Actual} \quad \mathcal{R} \quad \longrightarrow \quad \text{Phenotype}$$

Equation (2)

The dark component is therefore not an entity whose every phenotype is zero. Its electrical and optical expressions are assumed to be negligible, but its gravitational expression is nonzero. It requires no volume assigned to an individual particle, yet its stress energy density has a spatial distribution. This distinction makes invisibility, absence of particle volume, and presence of mass mutually consistent.

2 Research Question and Model Assumptions

2 1 Research question

This paper asks one question. If ordinary matter is interpreted conventionally while only dark matter phenomena are rewritten as the mass phenotype of twist stress preserved in space, can galaxy rotation, gravitational lensing, and early structure formation survive simultaneously within one model? The question does not presuppose that particle dark matter is absent. If twist stress is insufficient, an additional particle owner must be returned to the cost ledger.

2 2 Assumptions and scope of claims

Table 2 Model assumptions and claim grades

Ledger — Symbol · Assumption · Claim grade

Symbol: A1

Assumption: The conventional account of ordinary matter and the four forces is retained

Claim grade: Model boundary

Symbol: A2

Assumption: The dark sector Actual is distributed spatial stress

Claim grade: Model choice

Symbol: A3

Assumption: Matter and light couple universally to one metric

Claim grade: Condition for covariant completion

Symbol: A4

Assumption: The universe may have finite boundaryless topology but no particular topology is imposed

Claim grade: Topological hypothesis

Symbol: A5

Assumption: Time is a relation among change and records rather than an independent substance

Claim grade: Ontological interpretation

Symbol: A6

Assumption: The cosmological background and galactic response share a matched normalization

Claim grade: Conditional closure

A4 and A5 are not theorems that follow automatically from the physical calculation. A6 is also not uniquely derived from the principle of minimal computation. This paper states these assumptions explicitly and identifies both the calculations they enable and the observations under which they fail.

3 The Present and the First Twist

3 1 The interpretation that only the present exists

The model does not require the past and future to exist as physical locations alongside the present. What exists is the present state. The past is a record remaining in the present, and the future is a possible update not yet realized. Time can be defined relationally by comparing the change of a repeatable clock variable with a reference rate of change.

$$\Delta t_{\text{clock}} \equiv \frac{\Delta C}{R_C}$$

Equation (3)

This interpretation does not deny change. States are updated and records differ. Consciousness reads differences among records as a continuous experience, but consciousness is not required for the computation of the universe. The covariant field tau used below is read as a label that orders states rather than as a substantive flowing time. Because its normalization enters the action and produces a physical degree of freedom, however, the tension with a fully timeless ontology remains in the cost ledger.

3 2 First quantization and the first twist

A completely undifferentiated state contains no direction, position, particle, or record. The first quantization distinguishes possible states, and the first twist creates differences among paths and directions. On this reading, the Big Bang is not primarily an explosion into an external empty space. It is the first distinction and asymmetry that opened the relational network of the present universe.

undifferentiated → distinguishable → relational universe

Equation (4)

Equation 4 is an interpretation of origin, not a microscopic process established by observation. The topological group and quantum degrees of freedom in which the first twist arose remain open questions. The paper assumes that origin and then asks whether the response of the present universe is calculable.

4 Finite Boundaryless Topology and Bidirectional Twist

4 1 Local geometry and global topology

A space that is almost flat locally may still be closed globally. Finite and boundaryless means that the space has finite volume while no point is an edge or a privileged center. It also removes any requirement that space expand into something external. To separate local fields from global information, the one form A is decomposed into Hodge components.

$$A = d\phi + \delta B + h, \quad dh = 0, \quad \delta h = 0$$

Equation (5)

$$\oint_{\gamma} h = \Theta_{\gamma} \neq 0$$

Equation (6)

The term $d\phi$ is the local gradient response, δB is the rotational response, and h is the harmonic sector that cannot be removed by local

differentiation. If Θ_γ , the integral of h along a closed path γ , is nonzero, global gluing information remains even when it is invisible locally. Saying that every place is a gluing surface does not mean that a physical seam lies at every point. It means that no local patch is designated as the beginning or end of the universe.

A Mobius strip is a useful analogy for orientation reversal and gluing, but the strip itself has one boundary and therefore is not an exact boundaryless model of the universe. A physical candidate must be sought among closed three dimensional manifolds. This paper does not select a particular manifold and retains only the global information that can in principle be measured through holonomy.

4.2 Bidirectional twist

Left handed and right handed twists may coexist. Energy and total stress depend on the sum of the squared components rather than on the sign of direction, while orientation is recorded separately by their difference. Total stress may therefore remain nonzero even when the net directional bias is small.

$$\kappa^2 = \kappa_+^2 + \kappa_-^2, \quad \chi_T = \frac{\kappa_+^2 - \kappa_-^2}{\kappa_+^2 + \kappa_-^2}$$

Equation (7)

When χ_T is zero, the two directional stresses have equal magnitude. When its absolute value approaches one, one direction dominates. This parameter may be related to circularly polarized gravitational waves, parity odd correlations, or global directionality data, but this paper does not derive a specific signal amplitude.

4.3 Relationship between expansion and twist

What is observed is the redshift between galaxies, angular diameter distance, luminosity distance, and structure growth. The picture of space stretching outward is one interpretation of those relations. If the same distance relations can be reproduced by a changing global holonomy or gluing rule, expansion and twist may be observationally equivalent

descriptions. Renaming the description alone would not constitute new physics.

Table 3 Relationship between expansion and twist interpretations

Ledger — Feature · Expansion description · Twist description · Discriminating criterion

Feature: Background distance

Expansion description: Change of scale factor

Twist description: Change of relational gluing and holonomy

Discriminating criterion: Interpretively equivalent if identical

Feature: Lensing and slip

Expansion description: Calculated from metric potentials

Twist description: Stress response contributes to the metric

Discriminating criterion: Distinct slip can discriminate

Feature: Global topology

Expansion description: Curvature and topology supplied as inputs

Twist description: Residual twist along a closed path

Discriminating criterion: Repeated correlations and matched circles

Feature: Local boundary effect

Expansion description: Usually absorbed into local fields

Twist description: Possible nonlocal response of global stress

Discriminating criterion: Environment dependent RAR residuals

Interpretive boundary

The model therefore does not reject the observed distance relations. It interprets their Actual as changing gluing and stress rather than as an expanding container. Distinguishing the two descriptions physically requires lensing slip, global topological correlations, or a nonlocal response to boundary conditions.

4 4 Identifiability of the size of the universe

$$L_\gamma = |\Theta_\gamma| \frac{c}{H_0} \sqrt{\frac{\zeta}{6f_T}}$$

Equation (8)

Equation 8 shows that a representative closed path length L_γ can be calculated if the holonomy Θ_γ , stiffness ζ , and stress fraction f_T are known. Current observations, however, constrain only the product $\zeta \kappa^2$ rather than ζ and the twist rate κ separately. Calculating the size of the universe from twist alone therefore requires one additional independent measurement of stiffness or holonomy. Single values such as 73 Gly or 147 Gly are conditional results that require extra assumptions; they are not conclusions of this paper.

5 Spatial Stress and the Mass Phenotype

5 1 Minimal stress energy

A space with twist rate κ can store deformation energy. In the simplest quadratic response, the stress energy density is defined as follows. The dimensionless stiffness ζ is restricted to positive values for sign stability.

$$u_T = \frac{\zeta c^4 \kappa^2}{16\pi G}$$

Equation (9)

$$\alpha \equiv \frac{c\kappa}{H_0}, \quad f_T \equiv \frac{u_T}{\rho_c c^2} = \frac{\zeta \alpha^2}{6}$$

Equation (10)

The quantity ρ_c is the critical density. Once observation fixes the effective dark sector fraction f_T , the product $\zeta \alpha^2$ is fixed, but ζ and α remain inseparable. This degeneracy is not a calculation error. It identifies the combination that the data actually measure. Reporting stiffness and twist rate separately without another observation would overestimate what has been inferred.

5 2 Why stress appears as mass

Spatial stress need not carry electric charge or optical emission. If it modifies the metric and affects the paths of matter and light, however, observers translate that effect into a gravitational source. Moving the modified field response to the source side of the standard Poisson equation defines an effective dark density.

$$\rho_T^{\text{eff}} = -\frac{1}{4\pi G} \nabla \cdot [(1 - \mu)\mathbf{g}]$$

Equation (11)

The quantity ρ_T^{eff} is not a particle number density. It is a mass phenotype that expresses the same observational effect in the language of standard gravity. Because the mass of ordinary matter remains unchanged, the model does not claim that every mass in the universe is stress. Only the additional gravity classified as dark matter is reinterpreted as a stress Actual.

Table 4 Expression channels of matter and twist stress

Ledger — Observation channel · Ordinary matter · Twist stress

Observation channel: Electrical interaction

Ordinary matter: Appears through charge and current

Twist stress: Assumed to have no direct expression

Observation channel: Optical emission and absorption

Ordinary matter: Appears through electromagnetic states

Twist stress: Assumed to have no direct expression

Observation channel: Individual particle volume

Ordinary matter: Particle and composite descriptions apply

Twist stress: Not required

Observation channel: Spatial distribution

Ordinary matter: Distributed as matter density

Twist stress: Distributed as stress energy density

Observation channel: Gravity and lensing

Ordinary matter: Source of the metric

Twist stress: Appears as an effective mass phenotype

Background stress may exist throughout the universe, including our local environment, but it need not be perfectly uniform. If baryonic distributions, boundary conditions, and global holonomy modify the local response, the center of effective density may differ among galaxies and galaxy clusters. Computing this distribution is a central task for subsequent work.

6 Minimal Galactic Response

6 1 Static effective energy and field equation

The static weak field of a galaxy is described by a dimensionless potential ϕ and an acceleration scale a_T . Connecting the Newtonian and low acceleration regimes with one response function F can reduce the cost of assigning an independent halo profile to every galaxy.

$$E[\phi] = \int d^3x \left[\frac{a_T^2}{8\pi G} F(Y) + \rho_b c^2 \phi \right]$$
$$Y = \frac{c^4 |\mathbf{A}|^2}{a_T^2}, \quad \mathbf{g} = -c^2 \mathbf{A}$$

Equation (12)

$$\nabla \cdot \left[\mu \left(\frac{|\mathbf{g}|}{a_T} \right) \mathbf{g} \right] = -4\pi G \rho_b, \quad \mu = F_Y$$

Equation (13)

In the static limit, Equation 13 belongs to the same mathematical family as AQUAL or MOND type modified gravity [1,2]. This paper does not claim the equation itself as an original invention. Its distinct contribution is to interpret this response as an Actual of spatial stress and a mass phenotype, and to connect it with stress at cosmological scales.

6 2 Deep low acceleration regime

$$F(Y) = \frac{2}{3} Y^{3/2} \Rightarrow \mu(x) = x, \quad g = \sqrt{a_T g_b}, \quad v_f^4 = G M_b a_T$$

Equation (14)

Under spherical symmetry or a sufficiently isolated circular orbit approximation, Equation 14 produces the forms of both the observed radial acceleration relation and the baryonic Tully Fisher relation [3,4]. Because one a_T enters both relations, the computational cost is lower than assigning a separate set of dark halo parameters to every galaxy. Fits that include nonsphericity, gas distribution, external fields, and distance errors must nevertheless be tested through a hierarchical analysis of the full SPARC data set.

6 3 Static stability

$$\lambda_{\perp} = \mu > 0, \quad \lambda_{\parallel} = \mu + x\mu' > 0$$

Equation (15)

In the deep regime, $\mu = x$ gives the two positive eigenvalues x and $2x$. The ellipticity of the static field equation is therefore preserved. This does not prove that every mode of the full covariant theory is stable. Cosmological scalar modes, the ultra long wavelength Jeans sector, and nonlinear defects require separate tests.

7 Connecting Cosmological and Galactic Scales

7 1 Matched response condition

The model imposes a condition that the cosmological background stress and the galactic low acceleration response share the same normalization under a three dimensional isotropic projection. This is called matched response with $C = 1$. The cosmological Hubble combination and stress fraction then determine the galactic transition acceleration.

$$a_T = cH_0 \sqrt{\frac{r_T}{8}}, \quad f_T = 8 \left(\frac{a_T}{cH_0} \right)^2$$

Equation (16)

7 2 Numerical test

Table 5 Numerical substitution in the cross scale relation

Ledger — Input or output · Value · Meaning

Input or output: H_0

Value: $67.4 \text{ km s}^{-1} \text{ Mpc}^{-1}$

Meaning: Cosmological scale of relational change

Input or output: f_T

Value: 0.265

Meaning: Cosmological fraction of effective dark stress

Input or output: cH_0

Value: $6.55 \times 10^{-10} \text{ m s}^{-2}$

Meaning: Cosmological acceleration combination

Input or output: a_T

Value: $1.192 \times 10^{-10} \text{ m s}^{-2}$

Meaning: Galactic low acceleration transition scale

Input or output: Inferred f_T

Value: approximately 0.269

Meaning: Value obtained from $a_T = 1.20 \times 10^{-10} \text{ m s}^{-2}$

The calculated a_T is close to the empirical acceleration scale of approximately $1.2 \times 10^{-10} \text{ m s}^{-2}$ used in galactic dynamics [3]. Conversely, substituting that empirical value gives f_T of approximately 0.269. The meeting of quantities inferred at different scales within one relation is relevant to interpretive closure.

The value $f_T = 0.265$, however, is imported from an analysis performed within standard cosmology [5]. The agreement in Equation 16 is therefore not presented as a fully model independent prediction. It is a conditional derivation under the matched response assumption. Discriminating power requires a simultaneous fit to independent likelihoods from galactic and cosmological data.

8 Gravitational Lensing

8 1 One metric

For the mass phenotype to replace a conventional dark matter interpretation, it must affect the deflection of light as well as stellar motion. The model assumes that matter and light couple universally to one metric and uses two scalar potentials in the static weak field limit.

$$ds^2 = -\left(1 + \frac{2\Phi}{c^2}\right)c^2 dt^2 + \left(1 - \frac{2\Psi}{c^2}\right)d\mathbf{x}^2$$

Equation (17)

$$\hat{\alpha} = \frac{1}{c^2} \int \nabla_{\perp} (\Phi + \Psi) dl$$

Equation (18)

8 2 Lensing mass and dynamical mass

$$\Phi = \Psi + \mathcal{O}(c^{-2}) \Rightarrow \eta_{\text{slip}} = \frac{\Psi}{\Phi} = 1 + \mathcal{O}(c^{-2}), \quad M_{\text{lens}} = M_{\text{dyn}}$$

Equation (19)

If Equation 19 holds, twist stress is not a nonmetric correction that changes rotation curves while missing light. The same effective mass phenotype appears in material orbits and light deflection. The condition $\Phi = \Psi$ must nevertheless be established by the covariant completion. If anisotropic stress or additional degrees of freedom generate a large slip, agreement between lensing mass and dynamical mass may fail.

Colliding galaxy clusters provide a stronger test [6]. When the baryonic gas and lensing centers separate, the model must solve how the spatial stress field moves and persists dynamically. A static spherically symmetric equation cannot by itself explain that separation. Nonspherical lensing shear and postcollision centroid motion are priority numerical experiments for the model.

9 Early Universe and Structure Formation

9 1 Covariant computational structure

Carrying the static galactic response into the early universe requires a covariant action. This paper uses mathematics close to the relativistic Khronon theory of Blanchet and Skordis as computational material [7]. This section imports a covariant completion and is not an original derivation of that completion.

$$S = \frac{c^3}{16\pi G} \int d^4x \sqrt{-g} [R - 2J(Y) + 2K(Q)] + S_m$$

Equation (20)

$$n_\mu = -\frac{c}{Q} \nabla_\mu \tau, \quad Q = c \sqrt{-g^{\mu\nu} \nabla_\mu \tau \nabla_\nu \tau}$$

$$A_\mu = c^2 n^\nu \nabla_\nu n_\mu, \quad Y = \frac{A_\mu A^\mu}{c^4}$$

Equation (21)

The function J governs the spatial stress response in galaxies, while K governs the cosmological response that allows stress to cluster like an almost pressureless component. Tau may be read as a relational ordering label for states, but K gives physical meaning to its normalization Q. This covariant completion is therefore not identical to a purely reparameterization invariant and timeless ontology. The difference remains explicitly in the model cost.

9 2 DBI type response and minimum numerical test

$$K(Q) = \frac{2\mu^2}{\lambda_D} \left[1 - \sqrt{1 - \lambda_D(Q-1)^2} \right]$$

Equation (22)

$$q(a) = Q - 1 = [\lambda_D + (a^3/r)^2]^{-1/2}, \quad r = \frac{l_0}{2\mu^2}$$

Equation (23)

$$w = \frac{\lambda_D q^2 + (1+q) \sqrt{1 - \lambda_D q^2} - 1}{2 + (1 + \lambda_D)q}, \quad c_{ad}^2 = \frac{q(1 - \lambda_D q^2)}{1 + q}$$

Equation (24)

$$c_s^2(a, k) = c_{ad}^2 \left[1 + \frac{c_{ad}^2 k^2}{4\pi G a^2 \rho_T (1+w)} \right]^{-1}, \quad \Sigma_T = 0$$

Equation (25)

Using $H_0 = 67.4 \text{ km s}^{-1} \text{ Mpc}^{-1}$, a present stress fraction of 0.265, μ inverse = 22.3 Mpc, and $\lambda D = 1$, and normalizing the integration constant I_0 with the present density, gives $r = 9.9913 \times 10^{-6}$.

Table 6 Minimum numerical test of the DBI type response

Ledger — Regime · q · w · c ad squared

Regime: Very early a much less than 10⁻³

q: approaches 1

w: approaches 0

c ad squared: approaches 0

Regime: Recombination $a = 1$ divided by 1100

q: 0.999999997

w: 3.76×10^{-5}

c ad squared: 2.83×10^{-9}

Regime: Present $a = 1$

q: 9.99×10^{-6}

w: 5.00×10^{-6}

c ad squared: 9.99×10^{-6}

Regime: Relational variable $a = 10$

q: 9.99×10^{-9}

w: 5.00×10^{-9}

c ad squared: 9.99×10^{-9}

Near recombination, w and the adiabatic sound speed are very small, and the wavenumber dependent physical sound speed is smaller still. The anisotropic viscosity is zero. The component may therefore behave like cold dark matter in the background and in linear perturbations. This test

checks only a minimum survival condition for early structure formation. It does not replace a joint likelihood analysis of the full CMB angular spectrum, BAO, matter power spectrum, and growth rate data.

10 Measurability and Failure Conditions

10 1 What must be measured

Measuring only the mass phenotype makes a particle halo difficult to distinguish from spatial stress. Discrimination requires either a nonlocal response of the stress Actual or an environmental dependence that differs from a particle halo. If the local gravitational field responds when a global boundary condition B boundary changes, the following functional derivative may be nonzero.

$$\frac{\delta g(\mathbf{x})}{\delta B_{\text{boundary}}(\mathbf{y})} \neq 0$$

Equation (26)

No present technology is shown to relax spatial stress artificially, and no expected signal amplitude has been derived. Local stress release is therefore an experimental proposal rather than an experimental result. A more practical first step is to use natural differences in boundary conditions supplied by astronomical environments.

Table 7 Measurement paths that can distinguish the twist stress Actual

Ledger — Test · Observable · Requirement of the twist model · Status

Test: Galactic environment

Observable: RAR residuals and external field

Requirement of the twist model: Systematic change correlated with boundary conditions

Status: Testable

Test: Nonspherical lensing

Observable: Shear and dynamical mass

Requirement of the twist model: One stress distribution explains both

Status: Testable

Test: Colliding clusters

Observable: Gas stellar and lensing centroids

Requirement of the twist model: Dynamically reproduce motion and persistence of the stress field

Status: Open critical test

Test: Global topology

Observable: Repeated CMB correlations and matched circles

Requirement of the twist model: Observable trace of nonzero holonomy

Status: Open test

Test: Structure growth

Observable: CMB BAO σ_8 and power spectrum

Requirement of the twist model: Maintain small sound speed and allowed growth rate together

Status: Calculation required

Test: Local manipulation

Observable: Gravitational field change without inertial mass change

Requirement of the twist model: Direct response to a stress boundary condition

Status: Experimental proposal

10.2 Failure conditions

A minimal computation model must avoid failure under every declared condition. Success in one domain does not compensate for failure in another. If any of the following failures is structural and can be patched only by extra parameters, the present model must be rejected or replaced by a higher cost version.

- One response function fails to preserve both the RAR and the baryonic Tully Fisher relation across diverse galaxies.

- The covariant completion produces gravitational slip beyond observational limits and cannot explain lensing and dynamical masses together.
- In colliding galaxy clusters, the model cannot generate dynamics that separate the stress center from the observed lensing center.
- The effective sound speed or an unstable early universe mode damages both the CMB and structure growth.
- Reproducing observed distance relations in a finite boundaryless interpretation requires a separate gluing rule for each target.
- The covariant completion necessarily introduces an independent temporal degree of freedom that conflicts with the relational interpretation of time.

$$\text{Accept}_{2,2} \iff \text{survive}\{\text{galaxy, lensing, CMB, structure}\}$$

Equation (27)

11 Claim Grades and Position of the Paper

Table 8 Claim grade ledger for twist stress cosmology

Ledger — Claim · Grade · Exact meaning

Claim: WRRR Core 1.0

Grade: Preserved

Exact meaning: Not modified in this paper

Claim: Only the present exists

Grade: Ontological interpretation

Exact meaning: Time is interpreted as a relation among change and records

Claim: First twist

Grade: Origin interpretation

Exact meaning: The Big Bang is read as the opening of a relational network

Claim: Finite boundaryless universe

Grade: Topological hypothesis

Exact meaning: No specific three dimensional manifold has been selected

Claim: The dark Actual is spatial stress

Grade: Model choice

Exact meaning: Particle dark matter is not required as an owner

Claim: Dark mass is a phenotype

Grade: Theoretical redescription

Exact meaning: Modified geometry is observed as an effective density

Claim: Modified galactic response

Grade: Existing mathematics

Exact meaning: AQUAL and MOND type static limit

Claim: Cross scale relation

Grade: Conditional closure

Exact meaning: Matched response condition $C = 1$

Claim: Agreement of lensing mass

Grade: Leading order result

Exact meaning: Holds when $\Phi = \Psi$

Claim: DBI cosmology

Grade: Imported covariant completion

Exact meaning: Khronon mathematics is used as computational material

Claim: Full CMB fit

Grade: Open test

Exact meaning: Requires a Boltzmann code and likelihood analysis

Claim: Local stress release

Grade: Experimental proposal

Exact meaning: Implementation and signal amplitude remain unknown

The interpretive result of this paper is to join known phenomena within one Actual and Phenotype structure and to identify a minimum computation path without a fatal conflict across scales. A new discriminating prediction can strengthen the theory, but it does not by itself determine whether the interpretation is legitimate. Conversely, the absence of an independent prediction does not permit assumptions to be stated as facts. The claim grades guard against both errors.

12 Conclusion

Twist stress cosmology preserves the conventional account of ordinary matter and interprets only the additional gravity attributed to dark matter as the mass phenotype of stress stored in space. Electrical and optical expressions are assumed to be negligible, and no individual particle volume is required, but a gravitational expression exists because the stress changes the metric. Saying that the dark component is distributed throughout the universe, including our surroundings, means that a stress field has both background and local structure; it does not mean that particles fill space uniformly.

A Hodge decomposition of local fields and global topology permits a finite boundaryless universe to coexist with ordinary local gravitational fields. Bidirectional twist can preserve total stress while keeping net directional bias small. The Mobius strip is only an analogy for orientation reversal, not an exact boundaryless universe. A representative size of the universe cannot be reduced to one number until stiffness and holonomy are known independently.

The static low acceleration limit uses existing MOND type mathematics to recover the RAR and baryonic Tully Fisher relation. Under the matched response condition, cosmological H_0 and fT give $aT = 1.192 \times 10^{-10} \text{ m s}^{-2}$. With one metric and small slip, lensing mass and dynamical mass agree at leading order. A covariant DBI type response produces small pressure and sound speed near recombination, but full CMB and colliding cluster tests remain open.

This paper therefore does not declare a final theory superior to standard cosmology. Our universe is not special, and survival precedes any ranking

of best or worst. The present calculation shows that one owner, twist stress, may avoid failure in galaxies, lensing, and the early universe. The next steps are a joint SPARC fit, a Boltzmann implementation of covariant perturbations, stress dynamics in colliding clusters, and a search for global holonomy. If the model fails those tests, it must be discarded or revised with the added cost made explicit. This is the closure required by Minimal Computing Cosmology.

Part VII Interfaces to Standard Physics and Observational Calculation

2 Actual Renderer Reality Record

Version 2.3.2 does not mix ontology and calculation in a single sentence. Actual is the layer in which the common-carrier state, boundary relations, and twist exist. Renderer assigns the Actual state to measurable phenotypes such as particles, fields, mass, distance, and time. Reality is the layer of field equations and metrics used by experimenters. Record and Ledger preserve observations, sources, scopes of application, and update histories.

$$\Phi^R = \mathcal{R}[X_A], \quad \mathcal{O} = \mathcal{M}[\Phi^R]$$

Equation 2 Correspondence from Actual to Reality and observables

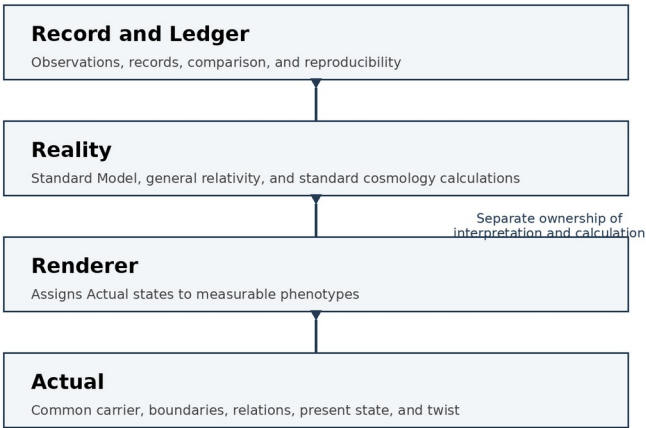


Figure 1 Four layers of Minimal Computing Cosmology and their information flow

Layer separation prevents the same object from being counted twice. The Standard Model electron and a WRRRA common-carrier state are not two electrons. The electron is a phenotype calculated in Reality, while the common carrier is the Actual basis from which that phenotype arises. Likewise, the dark-matter tensor and twist stress describe the same gravitational source in two languages and are not added simultaneously.

2 1 Ownership Rules

Ledger — Object · Reality calculation ownership · Actual interpretation ownership

Object: Ordinary matter particles and three forces

Reality calculation ownership: Standard Model

Actual interpretation ownership: Boundary-specific channel phenotypes of the common carrier

Object: Ordinary matter mass

Reality calculation ownership: Higgs coupling and QCD binding energy

Actual interpretation ownership: Phenotype of a state assigned as mass

Object: Gravity

Reality calculation ownership: Metric of general relativity

Actual interpretation ownership: Geometric transmission of total state change

Object: Dark matter gravitational source

Reality calculation ownership: Effective CDM tensor of Λ CDM

Actual interpretation ownership: Mass phenotype of local twist stress

Object: Redshift and distance

Reality calculation ownership: FLRW scale factor and light paths

Actual interpretation ownership: Distance and wavelength phenotypes of global twist

Object: Cosmic acceleration

Reality calculation ownership: Λ or a validated background module

Actual interpretation ownership: Background phenotype of global twist

3 Integration Rules

3 1 Module Adoption

An external theory becomes an official module of version 2.3.2 when it satisfies four conditions at once. First, it must have a domain validated by observation or experiment. Second, it must not logically contradict the fixed propositions of WRRRA. Third, ownership of what it calculates and at which layer must be explicit. Fourth, it must not duplicate a physical quantity already counted by another module.

$$M \in \mathfrak{M} \iff V(M) \wedge C(M) \wedge O(M) \wedge D(M)$$

Equation 3 Module-admission condition defined by validation V
consistency C ownership O and non-duplication D

Adopting a module does not import the entire ontology of that theory. Version 2.3.2 adopts its calculated results and validated domain, then reassigns their ontological meaning through the layer rules of WRRRA. Source, version, conditions of application, and output are stored in Record. Once this procedure is complete, integration is complete; re-derivation inside WRRRA is not obligatory.

3 2 Module Hierarchy

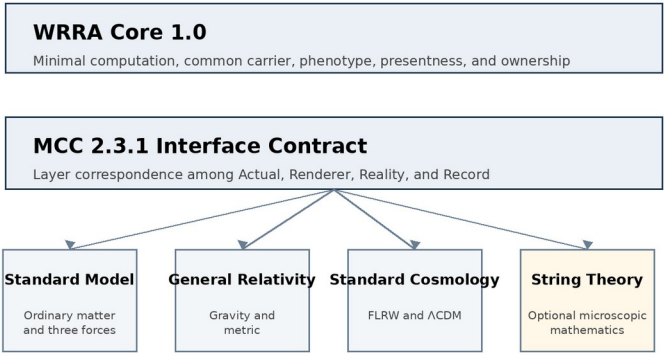


Figure 2 Hierarchy of WRRRA Core and adopted and optional modules

Ledger — Status · Module · Scope of use

Status: Higher specification

Module: WRRRA Core 1.0

Scope of use: Layers update minimal computation ownership phenotype

Status: Cosmological profile

Module: Minimal Computing Cosmology 2.3.2

Scope of use: Cosmological correspondence and integration

Status: Validated calculation

Module: Standard Model

Scope of use: Ordinary matter and strong weak electromagnetic forces

Status: Validated calculation

Module: General relativity

Scope of use: Metric gravity lensing orbits gravitational waves

Status: Validated calculation

Module: FLRW and Λ CDM

Scope of use: Background distances CMB and structure growth

Status: Optional mathematics

Module: String theory

Scope of use: Used only to implement common-carrier microstructure

3 3 What Integration Does Not Permit

- The common carrier is not treated as an additional particle beside the photon, gluon, W, and Z.
- An Actual state and its Reality particle tensor are not added simultaneously as gravitational sources.
- Extra dimensions and a particular vacuum from string theory are not automatically imported as established propositions of version 2.3.2.
- The numerical successes of standard cosmology are retained and are not refitted with arbitrary correction terms.
- The incompleteness of optional research is not recast as a failure to explain existing observations.

4 The Standard Model and Ordinary Matter

4 1 Adopted Scope

The Standard Model calculates interactions among quarks, leptons, gauge bosons, and the Higgs field within the $SU(3)_C \times SU(2)_L \times U(1)_Y$ gauge structure. Version 2.3.2 retains its calculations of the particle inventory, scattering amplitudes, coupling constants, mixing matrices, and symmetry breaking. Experimental values and uncertainties compiled through the Particle Data Group review system are adopted as inputs and validation records for the Reality module.

$$\mathcal{L}_{SM} = \mathcal{L}_{\text{gauge}} + \mathcal{L}_{\text{fermion}} + \mathcal{L}_{\text{Higgs}} + \mathcal{L}_{\text{Yukawa}}$$

Equation 4 Modular decomposition of the Standard Model Lagrangian

The masses of fermions and the W and Z bosons are calculated through their couplings to the Higgs field. Most of the mass of hadrons such as protons and neutrons comes not from the sum of quark rest masses but from QCD field energy and binding dynamics. The WRRRA mass phenotype does not erase this distinction. It interprets different generation mechanisms as being assigned through Renderer to the common measurable quantities of inertial and gravitational mass.

4 2 The Common Carrier and the Three Gauge Forces

The common carrier is not a fourth gauge boson. It is the common capacity in Actual that enables change to be transmitted, which Renderer expresses as different gauge interactions according to boundary conditions and channels. The gauge symmetries and field equations of the Standard Model are therefore retained. The unification of forces is not the forced merger of the three observed forces into a larger gauge group. It is a higher-order assignment in which different forces arise from the same transmission basis in Actual.

$$C_A \longrightarrow \{SU(3)_C, SU(2)_L, U(1)_Y\}_R$$

Equation 5 Generative correspondence from the common carrier to Standard Model gauge channels

Equation 5 does not claim to derive the particles and coupling constants of the Standard Model. It is a connection rule that preserves the WRRRA interpretation of matter developed from versions 1.0 through 2.1 while fixing the exact calculation ownership of Reality in the Standard Model. Direct derivation of particle species, generations, charges, and mixing angles from an Actual state remains optional microscopic research.

4 3 Quantum Fields and Vacuum

Quantum field theory in Reality calculates correlation functions, perturbative expansions, vacuum fluctuations, and scattering probabilities. WRRRA does not automatically interpret virtual particle pairs used as calculation terms as hidden events continually created and annihilated in Actual. A stable physical vacuum is a state with no observable flow of

particle events; a particle event enters Record only when an external field or collision produces real particles. This distinction does not restrict quantum-field calculations. It separates ownership of a computational representation from ownership of an existence event.

5 General Relativity and Gravity

In version 2.3.2 gravity does not disappear merely because it is described as a phenotype. In Reality, gravity is the actual response of the metric to total energy-momentum. Matter moves on that metric, and light follows its null geodesics. Because the model retains the coordinate covariance and local conservation laws of general relativity, it also retains calculations of Solar System orbits, gravitational lensing, gravitational waves, and strong gravitational fields.

$$G_{\mu\nu}[g^R] + \Lambda g_{\mu\nu}^R = \frac{8\pi G}{c^4} (T_{\mu\nu}^m + T_{\mu\nu}^\Theta)$$

Equation 6 Gravitational field equation in Reality with untwisted matter and radiation plus twist stress

In Equation 6, T_m denotes all contributions from matter and radiation other than the dark component and the cosmological-constant term. It includes the energy-momentum of baryons, electrons, photons, neutrinos, and Standard Model fields. The cosmological constant remains on the left-hand side and is not counted again in T_m or T_Θ .

The common carrier does not reduce gravity to particle exchange of the same kind as a gauge force. The three gauge forces and gravity share the generative fact that state change is transmitted in Actual, while retaining different effective equations in Reality. This explains why gravity and the three forces operate in one universe without forcing their calculations into a single equation.

5 1 Equivalence Principle and Conservation

The ordinary-matter tensor calculated by the Standard Model and the effective tensor of twist stress enter the right-hand side of the same metric equation. Their sum must be covariantly conserved. This is not an interpretive option; it is an interface contract that follows from adopting

general relativity. If twist stress does not satisfy it, the local-twist implementation must be revised.

6 Selective Adoption of String Theory

6 1 What Is Adopted

String theory uses one-dimensional objects and their vibrational modes instead of point particles and describes their coupling to spacetime fields through a worldsheet action. It provides a consistent mathematical structure that includes quantum gravity and tools for treating gauge and gravitational degrees of freedom together. Version 2.3.2 accepts this mathematics as a candidate microscopic implementation of the common carrier. Worldsheet boundaries, vibrational modes, chiral channels, anomaly cancellation, and compression of surplus degrees of freedom may be used to refine the 3+2 structure and rotor candidates studied in version 2.1.

$$S_P = -\frac{T_s}{2} \int d^2\sigma \sqrt{-h} h^{ab} \partial_a X^\mu \partial_b X^\nu G_{\mu\nu}(X)$$

Equation 7 Polyakov action of the optional string module

In Equation 7, X maps the worldsheet to background spacetime, h is the worldsheet metric, and G is the background metric. This action is not an established equation of the common carrier itself. To adopt a string module, a specific implementation must reproduce the particle species and symmetries of the Standard Model, agree with the low-energy limit of general relativity, and avoid double-counting additional degrees of freedom.

6 2 What Is Not Adopted

A particular number of dimensions, compactification, vacuum choice, or supersymmetric particle spectrum from string theory does not automatically become a proposition of version 2.3.2. Mathematics demonstrating internal consistency, such as the anomaly-cancellation result of Green and Schwarz, may be used. Promoting a specific microscopic implementation to a validated Reality module, however,

requires an observational correspondence. Its present status is optional mathematical material.

Ledger — String-theory element · Status in version 2.3.2 · Condition for adoption

String-theory element: Worldsheet and vibrational modes

Status in version 2.3.2: Available

Condition for adoption: When they simplify common-carrier microstructure

String-theory element: Chiral structure and anomaly cancellation

Status in version 2.3.2: Available

Condition for adoption: When Standard Model symmetries are preserved

String-theory element: Extra dimensions and compactification

Status in version 2.3.2: Implementation candidate

Condition for adoption: When they do not duplicate observable degrees of freedom

String-theory element: Supersymmetric spectrum

Status in version 2.3.2: Undetermined

Condition for adoption: When experimental correspondence is established

String-theory element: Claim of completed quantum gravity

Status in version 2.3.2: Not adopted

Condition for adoption: Requires a concrete low-energy correspondence and validation

7 A Finite Boundaryless Universe and Twist

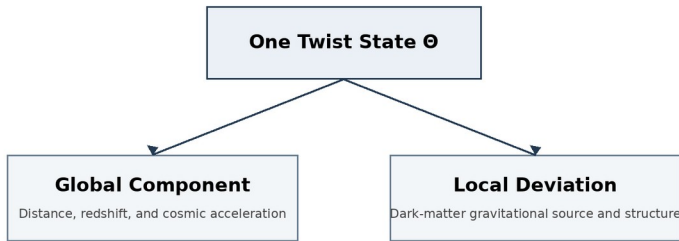
7 1 Definition of Twist

Twist does not mean that the universe is physically bent like a sheet of paper inside an external dimension. It denotes the residual relation of a global connection that remains different after a state is transported around

a closed path. A finite space without a boundary is possible in spherical spaces, tori, and quotient spaces. Version 2.3.2 does not fix the topology as a Möbius strip. The Möbius image is an analogy for directional reversal and nonprivileged gluing.

$$\Theta(x, \lambda) = \bar{\Theta}(\lambda) + \delta\Theta(x, \lambda)$$

Equation 8 Decomposition of one twist state into a global mean and local deviation



One state described by its mean and deviation, not two separate substances

Figure 3 Division of roles between global twist and local stress

The global component appears in Reality as the background of distance, redshift, and cosmic acceleration. Local deviations appear as additional gravitational sources and seeds of structure. They are not two separate substances, but the mean and deviation descriptions of one Actual state. In Equation 8, λ is a continuous approximation to the discrete updates n of Equation 1. Local twists with different directions are permitted when they satisfy total conservation and boundary conditions.

7 2 Total Content and Stress of the Universe

The total content of Actual is assumed not to leak outside. Ordinary matter retains its interpretation as existing fields and particles. Only the additional mass component corresponding to dark matter is interpreted as

the phenotype of local twist stress preserved in space. The total mass of the universe is not redefined as stress. This restriction preserves the Standard Model interpretation of matter and the lineage from versions 1.0 through 2.1.

8 The Mass Phenotype of Dark Matter

8 1 Tensor Correspondence

Dark-matter phenomena require a common additional gravitational source in galactic rotation curves, gravitational lensing, colliding clusters, the CMB, and structure growth. Version 2.3.2 does not deny this observational fact. It adopts the effective CDM tensor used by Λ CDM as a Reality calculation module and interprets the source of the same tensor in Actual as local twist stress.

$$T_{\mu\nu}^{\Theta} \equiv T_{\mu\nu}^{\text{CDM}}, \quad \nabla_{\mu}^R T_{\Theta}^{\mu\nu} = 0, \quad p_{\Theta} \simeq 0, \quad c_{s,\Theta}^2 \simeq 0, \quad \Pi_{\Theta} \simeq 0$$

Equation 9 Reality equivalence of the dark-matter tensor and twist stress and the conditions for inheriting CDM calculations

This equivalence does not add a separately renamed gravitational source. CDM and twist stress divide the calculation and interpretation ownership of the same term. To inherit the density-perturbation transfer functions and N-body results of standard CDM, twist stress must be separately covariantly conserved at the relevant scales; its effective pressure, sound speed, and anisotropic stress must be negligible; and its nongravitational coupling, dissipation, and self-interaction must remain below observational limits. If a self-action outside these conditions is introduced, it is tested as new dynamics rather than added simultaneously to the CDM module.

8 2 Colliding Clusters and Distribution

The weak-lensing map of the Bullet Cluster shows that the centers of the dominant baryonic plasma and the gravitational potential are separated. This observation requires an invisible additional gravitational source to move differently from the plasma during the collision. The twist-stress module must produce the same effectively nondissipative distribution as

the CDM tensor; otherwise that implementation fails. Observation does not directly detect the word twist, but it specifies the dynamics that the interpretation must reproduce.

8 3 Measurability of Local Stress Relaxation

To determine whether twist stress is more than a relabeling, one must search for metric effects left by the relaxation or redistribution of local stress. Candidates include time-delay lensing in colliding clusters, relaxation of lensing mass centers, phase offsets in nonspherical halos, and boundary-dependent residuals in precision gravimetry. Version 2.3.2 does not yet fix the magnitude of such effects. This measurement belongs to the observational-protocol branch of the four areas of further research.

9 A Twist Interpretation of Expansion and Redshift

9 1 FLRW Calculations Are Retained

Observers do not directly see galaxies fleeing into an external space. They measure relations among redshift, brightness, angular distance, and time delay. Standard cosmology calculates these relations with the FLRW metric and scale factor. Version 2.3.2 retains that calculation and interprets the scale factor a not as the exterior size of Actual but as the effective scale at which global twist appears in distance measurements in Reality.

$$ds_R^2 = -c^2 dt_R^2 + a_R^2(t_R) \left[\frac{d\chi^2}{1 - k\chi^2} + \chi^2 d\Omega^2 \right]$$

Equation 10 FLRW metric retained in Reality

$$1 + z = \frac{a_R(t_o)}{a_R(t_e)}, \quad a_R = \mathcal{R}_a[\bar{\Theta}]$$

Equation 11 Correspondence between standard redshift and global twist

$$H_R^2 = \left(\frac{\dot{a}_R}{a_R} \right)^2 = \frac{8\pi G}{3} \rho_R - \frac{kc^2}{a_R^2} + \frac{\Lambda c^2}{3}$$

Equation 12 Friedmann equation retained in Reality

In Equations 10 through 12, a_R is a scale factor with dimensions of length, χ and k are respectively a dimensionless coordinate and curvature index,

and k is one of -1, 0, or 1. The quantity ρ_R is the total effective mass density, including the contribution of radiation energy density divided by c^2 . When the energy density ε_R is used, the first term in the Friedmann equation is written as $8\pi G\varepsilon_R/(3c^2)$.

Equation 11 applies to cosmological redshift between a comoving emitter and observer in an FLRW background. An observed redshift may also include a Doppler component from peculiar velocity and a local gravitational redshift. In data analysis, these components are modeled or corrected separately before comparison with the background relation.

Expansion and twist use the same mathematics within their observationally equivalent domain, but they do not present the same ontological picture. The standard description defines distance through changes in a , while version 2.3.2 interprets those changes as updates of internal connections within one finite boundaryless Actual. Because no arbitrary redshift function is introduced, the model inherits existing supernova, BAO, and CMB fits.

9 2 A Conserved Actual and Changing Distances

$$\frac{dV_A}{d\lambda} = 0 \Rightarrow \frac{da_R}{d\lambda} = 0$$

Equation 13 Compatibility of a conserved-total marker for Actual with scale change in Reality

Equation 13 does not fix coordinate volume or the physical volume of Reality. V_A is a marker for the conserved total of Actual that assumes no exterior, whereas a_R represents distance relations compared by internal observers. The parameter λ continuously approximates the discrete updates of Equation 1. If gluing and connection are distributed across all positions and updated incrementally, systematic changes in internal distances are possible without a moving external boundary. The specific dynamics of this correspondence belong to subsequent interface research.

9 3 Cosmic Acceleration

In Reality, acceleration is calculated with the standard equations including the cosmological constant Λ . In Actual, the background state of global twist

appears as the acceleration phenotype. The mass phenotype of local stress and the acceleration phenotype of the global background need not be the same particle species. Components at different scales of the same twist state may be rendered as different effective terms.

10 The Big Bang and Presentness

10 1 The First Relational Opening

The Big Bang is not interpreted as the explosion of a point placed in empty space outside the universe. In version 2.3.2 it is the relational opening at which distinguishable states, relations, paths, and quantization first became established. The first twist may have appeared together with the first quantization that produced different paths and directions. This statement assigns meaning to the initial condition; it does not claim a completed equation of quantum gravity.

$$a_R \rightarrow 0 \not\Rightarrow V_A \rightarrow 0$$

Equation 14 Distinction between the initial limit in Reality and the total volume of Actual

In standard cosmology, the limit in which a approaches zero is an extrapolation toward a hot, dense initial state. Version 2.3.2 does not turn this into the proposition that the whole Actual vanished into a Euclidean point. Nucleosynthesis, recombination, the CMB, and primordial perturbations are calculated by the standard thermal-history module; the relational opening supplies the Actual meaning of those calculations.

10 2 Meaning of Observing the Past

To observe a galaxy from 13 billion years ago is to reconstruct, with the present metric, a record stored in light that arrives now. The past universe does not remain as a place outside the present universe. Every observation occurs in the present, while distance and lookback time are relations among records. This interpretation does not alter calculations using the speed of light, null geodesics, or redshift.

11 Integrated Calculation of Large Scale Structure

11 1 Inheritance of Standard Calculations

Primordial CMB perturbations and the growth of large-scale structure are treated with the linear perturbation and N-body calculations of standard cosmology. In version 2.3.2, the dark-matter density used in those standard calculations is read as the mass phenotype of twist stress. With the same initial conditions, metric, and effective tensor, the calculated filaments, walls, clusters, and voids are identical.

$$\mathcal{O}_{\text{MCC}}[X_A; \mathcal{R}] = \mathcal{O}_{\mathfrak{M}_{\text{calc}}}[\Phi^R], \quad \mathfrak{M}_{\text{calc}} = (\text{SM}, \text{GR}, \text{FLRW}/\Lambda\text{CDM})$$

Equation 15 Condition for observational equivalence between adopted modules and Minimal Computing Cosmology

In Equation 15, $\mathfrak{M}_{\text{calc}}$ is not a simple sum of the Standard Model, general relativity, FLRW, and ΛCDM . It is a composed calculation package in which each module has its own domain and input-output contract. Ownership of a physical quantity is assigned to only one module, and a calculated output is passed between modules as the next module's input.

11 2 Limited Structure Growth Calculation

The following two-dimensional toy calculation uses a 256×256 periodic grid and 65,536 mass elements. A Zel'dovich displacement is applied to one scalar perturbation field generated with fixed random seed 22447864. The rule is taken from a standard structure-growth approximation; only the reading of the scalar field as twist stress belongs to the interpretation of version 2.3.2. This is neither a cosmological parameter fit nor a calculation of an actual CMB transfer function.

Density Phenotype Growing under the Same Effective Gravitational Rule

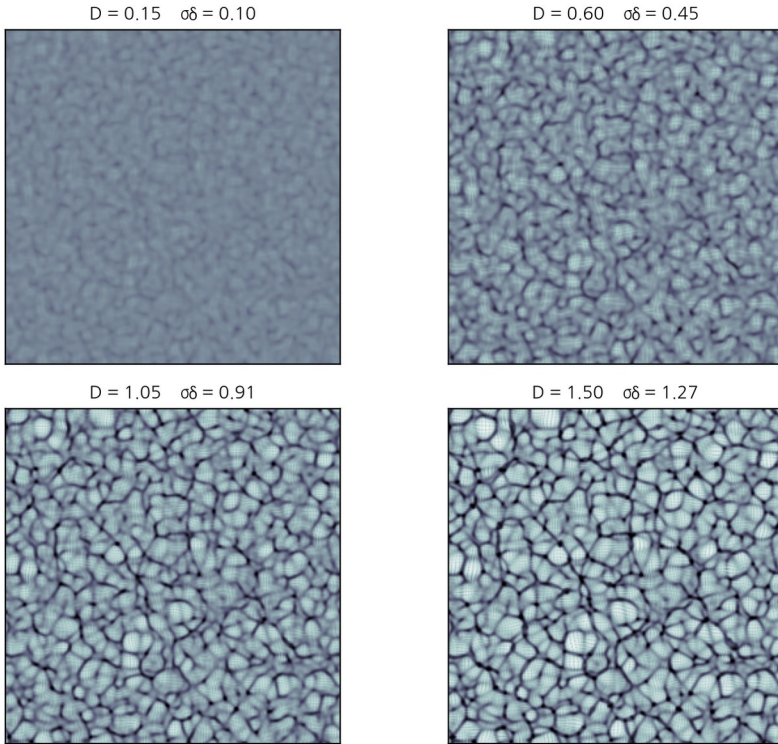


Figure 4 Density phenotype of twist stress growing under the adopted Zel'dovich rule

Ledger — Growth D · Density contrast standard deviation · Void fraction percent · High density fraction percent · Periodic connected components

Growth D: 0.15

Density contrast standard deviation: 0.103

Void fraction percent: 0.0

High density fraction percent: 0.0

Periodic connected components: 0

Growth D: 0.60

Density contrast standard deviation: 0.451

Void fraction percent: 6.0

High density fraction percent: 3.4

Periodic connected components: 416

Growth D: 1.05

Density contrast standard deviation: 0.915

Void fraction percent: 23.1

High density fraction percent: 8.4

Periodic connected components: 665

Growth D: 1.50

Density contrast standard deviation: 1.273

Void fraction percent: 37.3

High density fraction percent: 11.4

Periodic connected components: 856

As D increases from 0.15 to 1.50, the standard deviation of the density contrast rises from about 0.103 to 1.273, while the fraction of regions below half the mean density rises from 0 percent to about 37.3 percent. The mean-mass error remains at floating-point level. Connected components are counted with four-neighbor connectivity and periodic boundary conditions that identify the left-right and top-bottom edges. This result does not validate a new cosmology. It demonstrates implementability: structure-formation calculations are preserved under the twist-stress interpretation when the same effective gravitational rule is retained.

12 Full Inheritance of Observational Results

The successes of standard physics are not owned exclusively by a particular ontological vocabulary. Models using the same equations, initial conditions, boundary conditions, and observational correspondences produce the same numerical results within their shared domain. Version

2.3.2 makes this explicit and fixes the scope of inheritance in the following table.

Ledger — Observational domain · Adopted calculation · Actual interpretation in version 2.3.2 · Current judgment

Observational domain: Particle scattering and decay

Adopted calculation: Standard Model

Actual interpretation in version 2.3.2: Channel phenotype of the common carrier

Current judgment: Calculation inherited

Observational domain: Atoms and ordinary matter mass

Adopted calculation: Higgs QCD quantum field theory

Actual interpretation in version 2.3.2: Mass phenotype

Current judgment: Calculation inherited

Observational domain: Solar System and binaries

Adopted calculation: General relativity

Actual interpretation in version 2.3.2: Metric response

Current judgment: Calculation inherited

Observational domain: Gravitational waves

Adopted calculation: General relativity

Actual interpretation in version 2.3.2: Geometric transmission of Actual change

Current judgment: Calculation inherited

Observational domain: Galaxy rotation and lensing

Adopted calculation: GR plus CDM

Actual interpretation in version 2.3.2: Local twist stress

Current judgment: Calculation inherited

Observational domain: Colliding clusters

Adopted calculation: Nondissipative CDM tensor

Actual interpretation in version 2.3.2: Mobile stress distribution

Current judgment: Conditionally inherited

Observational domain: Redshift supernovae BAO

Adopted calculation: FLRW and Λ

Actual interpretation in version 2.3.2: Global-twist distance phenotype

Current judgment: Calculation inherited

Observational domain: CMB

Adopted calculation: Linear perturbations and recombination

Actual interpretation in version 2.3.2: Record of initial twist

Current judgment: Calculation inherited

Observational domain: Large-scale structure

Adopted calculation: Perturbations and N-body

Actual interpretation in version 2.3.2: Mass phenotype of stress distribution

Current judgment: Calculation inherited

Observational domain: Early thermal history

Adopted calculation: Nucleosynthesis and standard cosmology

Actual interpretation in version 2.3.2: Reality after relational opening

Current judgment: Calculation inherited

The final Planck mission data showed strong agreement with the spatially flat six-parameter Λ CDM model. Version 2.3.2 does not replace those parameters or the CMB acoustic scale with new values. Even when current data do not uniquely choose between the standard description and the ontology of version 2.3.2, the compatibility of the same calculations with version 2.3.2 forms the observational basis of the integrated model.

The 2025 DESI DR2 BAO analysis reported a preference for a time-varying dark-energy equation of state over Λ CDM in extended models combining DESI and CMB data. That preference remained after adding the 2026 Lyman-alpha forest full-shape analysis, although the central value of the

new high-redshift measurement moved toward Λ CDM. Version 2.3.2 therefore does not make Λ a fixed ontological conclusion. It retains Λ , or the latest validated background module, as the owner of the Reality calculation. Even when observational modules are updated, assignment under the Core to global twist as a distance and background phenotype remains unless a contradiction is demonstrated.

13 Restricted Further Research

Further research is opened only when the integrated model demonstrably fails to explain, calculate, or map an item, or when a logical contradiction or double-counting is found between adopted modules. Its scope is limited to the missing element and its interface. Recalculating under the WRRR name a phenomenon already calculated by the Standard Model, general relativity, or Λ CDM is not a required task.

13 1 Twist Metric Interface

Specify in one action or update equation the Renderer from global twist to a_R and H_R and the tensor rule from local twist to $T\Theta$. This research is needed when an independent dynamical calculation is desired. It is not a condition for the observationally equivalent interpretation of version 2.3.2.

13 2 Common Carrier Standard Model Interface

Study how a common-carrier state produces the representations of $SU(3)_C \times SU(2)_L \times U(1)_Y$, particle generations, charge, mass, and mixing. String worldsheets, a 3+2 internal structure, chiral channels, and a rotor are candidate tools. They remain optional implementations until a candidate actually reproduces the Standard Model.

13 3 Topology Presentness Interface

For admissible finite boundaryless topologies, calculate the global holonomy, total size, and range of directional reversal. At the same time, formalize the relation of the present update order λ to relativistic covariance and proper time. Both questions must be treated without reintroducing an exterior of the universe or an absolute clock.

13 4 Measurement Falsification Interface

Quantify residuals that local stress relaxation could leave in lensing time delays, colliding-cluster halo shapes, or precision gravimetry. Constrain global topology with repeated images, matched circles, and distance correlations. If the results are exactly identical to standard predictions, the proposal remains an ontological interpretation. It becomes an independent prediction only when it produces a reproducible difference.

Ledger — Research area · Opening condition · Protected component · Completion output

Research area: Twist metric

Opening condition: Need to calculate intrinsic evolution without standard modules

Protected component: FLRW and GR observational calculations

Completion output: Covariant action and Renderer

Research area: Common-carrier matter

Opening condition: Attempt to derive particle species and couplings

Protected component: Validated values of the Standard Model

Completion output: Gauge-preserving microscopic map

Research area: Topology and presentness

Opening condition: Attempt to formalize total size and λ

Protected component: Finite boundarylessness and proper-time calculations

Completion output: Topology group and covariant update rule

Research area: Measurement and falsification

Opening condition: A residual distinct from standard results is calculated

Protected component: Full inheritance of existing observations

Completion output: Preregistered statistical test

14 Failure Conditions and Order of Revision

Version 2.3.2 does not alter the Core in response to every objection. When failure is established, revision begins with the lowest-level implementation. Inputs and numerical implementation of the observational-fit module are checked first, followed by Renderer and the correspondence definitions. Local or global twist implementations are revised next. WRR Core is reopened by separate agreement only when failure follows directly from Core grammar.

- Revise the global Renderer if one global-twist correspondence is shown unable to reproduce redshift, distance, the CMB, and structure growth within the same FLRW metric.
- Revise the local-stress implementation if its tensor cannot satisfy conservation, lensing dynamics, and the nondissipative behavior of colliding clusters together.
- Discard a microscopic common-carrier implementation if its Renderer cannot preserve Standard Model gauge symmetries without duplicate degrees of freedom.
- Revise the topological hypothesis if finite boundarylessness and nonprivileged gluing cannot agree with admissible cosmic topologies and observational constraints.
- Reconsider the physical profile of presentness if present-state updating and relativistic covariance cannot be logically compatible.

14 1 Module Replacement Rule

If a new validated theory calculates the same phenomenon more accurately over a wider domain than an existing module, the Reality module may be replaced. Replacement does not alter WRR Core. Inputs, outputs, units, conservation laws, and observational correspondences are specified, and differences from the previous module are stored in Record. This structure allows version 2.3.2 to absorb future research without losing its own criteria.

15 Final Integrated Propositions

Ledger — No · Final proposition

No: 1

Final proposition: WRRRA Core 1.0 is the higher-order execution grammar of Minimal Computing Cosmology 2.3.2

No: 2

Final proposition: Our universe need not be special or optimal and maintains viable sufficiency

No: 3

Final proposition: The universe is one finite boundaryless present whole

No: 4

Final proposition: Time is a relation of change and record rather than an independent substance and Reality retains proper-time calculations

No: 5

Final proposition: Ordinary matter and the three gauge forces are calculated by the Standard Model in Reality

No: 6

Final proposition: Ordinary matter is interpreted in Actual as boundary-specific phenotypes of the common carrier

No: 7

Final proposition: Gravity is the actual metric response to total energy-momentum and is calculated by general relativity

No: 8

Final proposition: Dark matter is the mass phenotype of local twist stress and is not added again to the effective CDM tensor

No: 9

Final proposition: Expansion redshift and cosmic acceleration are distance and background phenotypes of global twist

No: 10

Final proposition: The Big Bang is the first relational opening rather than a point explosion in external space

No: 11

Final proposition: The Standard Model general relativity and standard cosmology are adopted as validated calculation modules

No: 12

Final proposition: String theory is an optional mathematical module used only when needed for a microscopic implementation of the common carrier

No: 13

Final proposition: External research is adopted without internal re-derivation when it satisfies validation consistency ownership and non-duplication

No: 14

Final proposition: Further research is limited to WRRR interfaces where a real gap or module conflict has been identified

Conclusion

Minimal Computing Cosmology 2.3.2 neither discards the Standard Model nor elevates it into the final ontology of the universe. The Standard Model calculates ordinary matter, general relativity calculates gravity, and standard cosmology calculates the background and structure formation. WRRR organizes these results as phenotypes of Reality produced through Renderer by one common-carrier and twist state.

Through this integration, the major known observations of the universe fall within the calculation scope of the model. Ordinary matter, dark matter, gravity, redshift, cosmic acceleration, the CMB, the early universe, and large-scale structure can already be calculated by adopted modules. The remaining research is not a list of unexplained cosmic properties. It is a list of interfaces required for WRRR to produce those calculations directly.

The originality of version 2.3.2 does not lie in discarding existing numerical results and fitting new ones. It reads reality as phenotype, expansion as the distance phenotype of global twist rather than the motion of an external boundary, and force unification as the generative basis of a common carrier rather than a single additional particle. This interpretation is logically compatible with standard calculations and closes as one conceivable finite boundaryless universe.

Part VIII Cumulative Integration and Nonduplicating Interfaces

Chapter 28 Native Twist Dynamics and the CDM Equivalence Branch

MCC 2.2 and 2.3.1 are not competing versions. They operate at different computational depths. In the native branch of 2.2, a twist action directly produces an effective tensor.

$$T_{\mu\nu}^{\Theta} = -\frac{2}{\sqrt{-g}} \frac{\delta S_{\Theta}}{\delta g^{\mu\nu}}$$

In the equivalence branch of 2.3.1, the same tensor is identified with the CDM computational representation within a declared scale and accuracy.

$$T_{\mu\nu}^{\Theta} \equiv T_{\mu\nu}^{CDM}, \nabla^{\mu} T_{\mu\nu}^{\Theta} = 0, p_{\Theta} \simeq 0, c_{s,\Theta}^2 \simeq 0, \Pi_{\Theta} \simeq 0$$

They are not entered twice in one gravitational equation.

$$G_{\mu\nu} + \Lambda g_{\mu\nu} = \frac{8\pi G}{c^4} (T_{\mu\nu}^m + T_{\mu\nu}^{\Theta}), T^{\Theta} \neq T^{\Theta} + T^{CDM}$$

The native branch gains its own calculation ownership when it passes conservation, lensing, colliding-cluster, CMB, and structure-formation tests. Where that calculation is not yet closed, the equivalence branch inherits validated calculations. This interface preserves twist as more than a renamed source while retaining successful standard computations.

Chapter 29 One Global and Local Twist State

$$\Theta(x, \lambda) = \dot{\Theta}(\lambda) + \delta\Theta(x, \lambda)$$

The global mean $\dot{\Theta}$ is rendered as a_R, H_R , and cosmological distance relations. The local deviation $\delta\Theta$ is rendered as $T_{\mu\nu}^{\Theta}$ and as the seed of structure growth.

$$a_R = R_a[\dot{\Theta}], T_{\mu\nu}^{\Theta} = R_T[\delta\Theta]$$

These are not two independent dark substances. They are a scale decomposition of one Actual twist state. The global mean is not added again to the dark-matter tensor, and the local deviation is not duplicated in the cosmological-constant term.

Chapter 30 Final Interface of the Common Carrier and Four Forces

The common carrier is neither a fifth force nor a field that flattens all four forces into one gauge current. A shared Actual transport basis is rendered through different owner maps as the three gauge representations and gravity.

$$R_{gauge}[C_A] \rightarrow \{SU(3)_C, SU(2)_L, U(1)_Y\}_R, R_G[C_A, \Phi_R] \rightarrow g_{\mu\nu}$$

The gauge forces selectively read J_a , while gravity reads total $T_{\mu\nu}$. The four forces are unified within one transport basis and one Reality precisely without being declared identical. The singlet counterexample and fifteen-channel equations of MCC 2.0 provide the argument; the Standard Model and general relativity modules of 2.3.1 provide the Reality calculations.

Chapter 31 Compatibility of a Matter Occupied Reality and the Twist Universe

Ordinary matter retains its conventional particle and field description. Only the additional gravitating dark component is read as a local twist-stress phenotype. At the same time, the lawful matter and antimatter sectors are separated from long-term actual occupancy.

$$H_{lawful} = H_{matter} \oplus H_{antimatter}, OccupiedReality(\lambda) \subset H_{matter}$$

Adding twist does not convert all ordinary matter into spatial stress. A matter-occupied Reality is a proposition about long-term ordinary-matter occupancy. The dark mass phenotype is a proposition about spatial stress that has negligible electric and optical expression but a nonzero gravitational expression. Their ownership domains differ and they are mutually compatible.

Chapter 32 The Present Universe and Experienced Time

Actual contains an updating present. The past remains as Record in the present, and the future is an unrealized possibility. Proper time in general relativity and the FLRW time coordinate remain effective quantities that calculate relations among change and records in Reality. Consciousness is not a prerequisite for cosmic computation. It is a phenotype that integrates memory and prediction into an experienced flow of time. Change proceeds without consciousness, but no subject experiences its duration.

When stability fails, a second universe is not created outside the first. The single Actual updates into a next state. An unstable state updates again, while a stable state persists for the duration of its stability. Finite boundarylessness means that no external edge is required; it does not require an infinite material container.

Chapter 33 Final Integrated Propositions

Ledger — Number · Fixed proposition of Minimal Computing Cosmology 2.3.2

Number: 1

Fixed proposition of Minimal Computing Cosmology 2.3.2: Our universe need not be privileged or optimally designed and is one persistent viable structure

Number: 2

Fixed proposition of Minimal Computing Cosmology 2.3.2: Minimal computation is the nearest sufficient computation and reserve that avoids fatal failure under change and extremes

Number: 3

Fixed proposition of Minimal Computing Cosmology 2.3.2: LAW SOURCE EXECUTION RENDERER REALITY and RECORD form one cosmic execution genealogy

Number: 4

Fixed proposition of Minimal Computing Cosmology 2.3.2: One constraint-preserving opening starts one history and radiation can arise during execution without a separate SOURCE register

Number: 5

Fixed proposition of Minimal Computing Cosmology 2.3.2: Gravity reads total stress energy while three gauge forces read selective currents so the four forces are asymmetrically unified

Number: 6

Fixed proposition of Minimal Computing Cosmology 2.3.2: Fifteen chiral channels share one common carrier skeleton while abstract rank and gauge-covariant realization remain distinct

Number: 7

Fixed proposition of Minimal Computing Cosmology 2.3.2: Matter is a jointly owned phenotype of mass charge spin location and record

Number: 8

Fixed proposition of Minimal Computing Cosmology 2.3.2: Antimatter may remain lawful while long-term occupied Reality consists of matter

Number: 9

Fixed proposition of Minimal Computing Cosmology 2.3.2: Quantum possibilities are weighted within lawful support and one event and Record condition the next present

Number: 10

Fixed proposition of Minimal Computing Cosmology 2.3.2: The universe may be a finite boundaryless Actual with no privileged edge or origin point

Number: 11

Fixed proposition of Minimal Computing Cosmology 2.3.2: First quantization and first twist interpret the Big Bang as the opening of relation and direction

Number: 12

Fixed proposition of Minimal Computing Cosmology 2.3.2: Global twist is rendered as distance redshift and background acceleration while local twist is rendered as dark mass and structure

Number: 13

Fixed proposition of Minimal Computing Cosmology 2.3.2: The native twist branch of 2.2 and the CDM equivalence branch of 2.3.1 are non-substitutive and are never double counted

Number: 14

Fixed proposition of Minimal Computing Cosmology 2.3.2: The Standard Model GR FLRW and validated cosmology are Reality modules and do not replace the WRRR skeleton

Number: 15

Fixed proposition of Minimal Computing Cosmology 2.3.2: String theory is optional microscopic material and does not delete minimal computation the common carrier or phenotype

Number: 16

Fixed proposition of Minimal Computing Cosmology 2.3.2: Past and future need not exist as separate places and time is a relation among change and records

Conclusion

The center of Minimal Computing Cosmology 2.3.2 is not a new name for dark matter. It is the full execution architecture in which the least independent input and reusable laws update the present, a common carrier and selective renderer produce dimensions, matter, forces, and observable Reality, and quantum possibility becomes one event and one recorded history.

The four forces are unified without being made identical. Gravity reads the $T_{\mu\nu}$ of the whole realized system, whereas electromagnetism, the weak force, and the strong force selectively read internal representation currents J_a . The common carrier preserves that distinction while

providing one Actual transport basis. Matter is not a stored list of numbers but a phenotype in which mass, charge, spin, location, and history are jointly owned. LAW may permit both matter and antimatter while a long-lived actual universe occupies the matter sector.

The twist universe is not a decorative later addition. It is the cosmological model that closes finiteness, dark gravity, and distance relations on top of the execution skeleton. Global holonomy and local stress are the mean and deviation of one state, rendered respectively as the distance phenotype of expansion and acceleration and the mass phenotype of dark matter. Native twist dynamics connects galaxies, lensing, and the early universe directly; the CDM-equivalence branch inherits validated structure-formation calculations under the same effective tensor conditions.

The Standard Model, general relativity, FLRW, and cosmological data enter as computational modules inside this skeleton. They are neither discarded nor promoted into the ontology of the whole architecture. They calculate where they have demonstrated accuracy, while WRRRA explains how their outputs arise from one Actual and become Reality and Record. Version 2.3.2 is therefore the cumulative integrated edition that preserves both the achievements of MCC 2.0 and the complete twist structure of MCC 2.2.

Appendix Core Equations and Claim Grades

Appendix A Ledger of Core Equations

Ledger — Function · Equation

Function: Overall execution

Equation: $LAW \rightarrow SOURCE \rightarrow EXECUTION \rightarrow RENDERER \rightarrow REALITY \rightarrow RECORD$

Function: Minimal cost

Equation:

$C_{total}=L_{LAW}+L_{SOURCE}+C_{read}+C_{evolution}+C_{render}+C_{record}$

Function: Opening operation

Equation: $U_S=\exp(i\eta\Omega/\hbar)$, $U_S^\dagger P_\Omega U_S=P_\Omega+\eta$

Function: FLRW boundary

Equation: $B_\eta=\eta\chi-\sqrt{6}\bar{M}_P|\eta|\ln(a/a^*)+constant$

Function: Dimension filter

Equation: $K_{DF}=F^\dagger F$, $L_{DF}=I-K_{DF}$

Function: Common transport

Equation: $D_C:\oplus H_i\rightarrow\oplus H_i$, $rank(D_C|_{band})=15$

Function: Gravity

Equation: $G_{\mu\nu}+\Lambda g_{\mu\nu}=(8\pi G/c^4)T_{\mu\nu_total}$

Function: Gauge coupling

Equation: $D_\mu=\nabla_\mu^{geo}-ig_1YB_\mu-ig_2T^aW_\mu^a-ig_3t^AG_\mu^A$

Function: Reality state

Equation: $\rho_{Reality}=P_{vis}[U(\tau)U_{Sp_dorm}U_S^\dagger U^\dagger(\tau)]$

Function: Event probability

Equation: $p(a|\rho)=Tr[I_a(\rho)]$

Function: Path equivalence

Equation: $R_{\{\gamma_1\}}(B_f,B_i)\simeq_{gauge} R_{\{\gamma_2\}}(B_f,B_i)$

Function: Recovery

Equation: $\text{Recovery} \circ \Phi = \text{id}$ on $A_{\text{protected}}$

Function: Support set

Equation: $\text{supp } P_{\text{WRRAC}} \subseteq \Gamma_{\text{lawful}}$

Function: History information

Equation: $I_{\text{history}}(T) = \Sigma_t - \log_2 p(a_t | h_{\{t-1\}})$

Function: Finite runtime

Equation: $H_N = H_{\text{Jacobi}} \otimes H_{\text{loc}} \otimes H_{\text{syn}}$

Function: Radiation generation

Equation: $\rho_r(t) = a(t)^{-4} \int_{[t^*]}^t a(t')^4 Q_r(t') dt'$

Function: Radiation takeover

Equation: $R = \rho_r / \rho_\chi$, $\dot{R}/R = 2H + Q_r(1/\rho_r + 1/\rho_\chi) \geq 2H$

Function: Matter asymmetry

Equation: $\Delta_M = n_M - n_{\bar{M}}$, $d(a^3 \Delta_M)/dt = a^3(S_M - S_{\bar{M}})$

Function: CP boundary

Equation: $F_B(\eta, \text{LAW}_{\text{CP-symmetric}}) = 0$

Appendix B Sources and Grades of Claims

The purpose of this ledger is to prevent established equations of standard physics from being mistaken for new derivations by WRRR, while also preventing WRRR's theoretical achievements from being buried under excessive qualification.

Ledger — Label · Meaning · Rule of Use

Label: BASELINE

Meaning: Established equations or observations of standard physics

Rule of Use: State the source and domain of applicability

Label: DEFINITION

Meaning: Layers or ownership assignments newly defined by WRRR

Rule of Use: Distinguish definitions from their consequences

Label: THEORETICAL RESULT

Meaning: Structural conclusion logically obtained from assumptions

Rule of Use: State the assumptions and the scope of counterexamples

Label: CONSTRUCTED/EXACT

Meaning: Exactly constructed or calculated in a specified model

Rule of Use: Do not identify it with the entire theory

Label: CALIBRATED

Meaning: Fixed using observational values

Rule of Use: Do not call it an independent derivation

Label: PREDICTION

Meaning: Output not used in calibration

Rule of Use: Freeze the procedure, timing, and failure conditions

Label: OPEN

Meaning: A connection not yet determined

Rule of Use: Maintain it as a separate list after the core achievements

Appendix C Major Public Research Genealogy

- 10.5281/zenodo.22038142 - Hypothesis on Gravity and Three-Dimensional Space
- 10.5281/zenodo.22052565 - Common-Carrier Unification without a Grand-Unified Gauge Field
- 10.5281/zenodo.22056516 - Minimal Common Transporter for Chiral Matter
- 10.5281/zenodo.22063551 - Autonomous Rank Opening of 15 Channels
- 10.5281/zenodo.22121047 - Conditional BRST Closure of the Common Carrier
- 10.5281/zenodo.22124886 - Renderer Path Independence and the Einstein Branch
- 10.5281/zenodo.22125506 - Recoverable Rendering and Metric Faithfulness
- 10.5281/zenodo.22126125 — time-uniform recoverable rendering
- 10.5281/zenodo.22126918 — finite-Jacobi compatibility benchmark
- 10.5281/zenodo.22137993 — open-universe execution model
- 10.5281/zenodo.22139233 — WRRRA information provision ledger
- 10.5281/zenodo.22141639 — two-pass open-outcome rendering
- 10.5281/zenodo.22142880 - The Position of WRRRA among Four Axiomatic Frameworks
- 10.5281/zenodo.22177447 - Final Our-Universe Scenario of Minimal Computing Cosmology
- 10.5281/zenodo.22239485 - Ownership of the Four Forces
- 10.5281/zenodo.22239574 - Big Bang, Dark Matter, and the Conservation Ledger
- 10.5281/zenodo.22239864 / 22239895 - Antimatter Direction-History Verification Program

- 10.5281/zenodo.22330256 / 22331940 - Information Closure and the Interpretation-Prediction Methodology
- 10.5281/zenodo.22447864 - A Physically Auditable Classifier of Interpretation and Prediction
- 10.5281/zenodo.22502619 - WRRRA-Worm Execution and Comparative Study

Appendix String Reinforcement Ledger

Appendix A. Claim-Grade Ledger

Ledger — Item · Grade · Exact Meaning

Item: Three invariants of MCC 2.0

Grade: CORE / PRESERVED

Exact Meaning: Must not be changed by this combination

Item: 10D superstring requirement

Grade: IMPORTED BASELINE

Exact Meaning: Microscopic consistency of perturbative superstring theory

Item: E8×E8 heterotic as primary material

Grade: MODEL CHOICE

Exact Meaning: Preferred candidate under MCC cost; no uniqueness claim

Item: 10D heterotic low-energy action

Grade: IMPORTED BASELINE

Exact Meaning: Not a new MCC derivation

Item: Green–Schwarz anomaly closure

Grade: IMPORTED EXACT/KNOWN

Exact Meaning: Existing result of string theory

Item: $D_C(4)=P15$ Red[$\mathcal{D}10$] P15

Grade: MCC PROPOSED IDENTIFICATION

Exact Meaning: Requires validation in a concrete compactification

Item: index = 3 target

Grade: CONDITIONAL

Exact Meaning: Not an automatic prediction of three generations

Item: String zero mode → phenotype

Grade: MCC THEORETICAL RECAST

Exact Meaning: Places string material in the upstream phenotype layer

Item: Qr from modulus/gauge kinetic function

Grade: CANDIDATE CONSTRUCTION

Exact Meaning: Numerical calculation remains OPEN

Item: ηB source from string sector

Grade: CANDIDATE

Exact Meaning: Mechanism selection and calculation remain OPEN

Item: WRR boundary $\leftrightarrow$ string microstates

Grade: CONDITIONAL IDENTIFICATION

Exact Meaning: Test beginning with specific black holes

Item: MCC landscape Pareto filter

Grade: NEW SELECTION PRINCIPLE

Exact Meaning: Requires actual vacuum ranking

Appendix B. Allowed and Forbidden Repair Material

Ledger — Decision · Material / Action · Reason

Decision: Allowed

Material / Action: 10D heterotic effective action

Reason: Microscopic gravity/gauge/anomaly owner

Decision: Allowed

Material / Action: Calabi–Yau/bundle compactification

Reason: Generates 4D chiral zero modes and representations

Decision: Allowed

Material / Action: Green–Schwarz mechanism

Reason: Reinforces anomaly consistency

Decision: Allowed

Material / Action: Gauge kinetic function / moduli couplings

Reason: Candidate microscopic owner of Qr

Decision: Allowed

Material / Action: String axion/moduli/hidden sectors

Reason: Dark-sector candidates

Decision: Allowed

Material / Action: Controlled black-hole microstate counting

Reason: Reinforces microscopic capacity of the WRRR boundary

Decision: Conditional

Material / Action: Type II/D-brane/F-theory material

Reason: Use when it has lower Pareto cost than heterotic material or supplies a required structure

Decision: Forbidden

Material / Action: Promoting 11D M-theory to the required skeleton

Reason: Outside the present scope and not required by this construction

Decision: Forbidden

Material / Action: Treating the string landscape itself as the selection explanation

Reason: Cannot be adopted without MCC minimality screening

Decision: Forbidden

Material / Action: Replacing the common carrier with the phrase “everything is string”

Reason: Eliminates the actual transport operator

Decision: Forbidden

Material / Action: String mode = observed particle

Reason: Destroys the phenotype layer

Decision: Forbidden

Material / Action: Extra SOURCE hiding radiation/asymmetry in vacuum labels

Reason: Violates minimal SOURCE

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Appendix Integrated Definitions and Research Lineage

Definitions

Ledger — Term · Definition

Term: WRRRA

Definition: Higher-order execution grammar organizing state relation law boundary update phenotype and record

Term: Minimal computation

Definition: Principle of maintaining enough computation and reserve to avoid fatal failure under changing conditions

Term: Actual

Definition: Existence layer containing the common carrier boundary relations present state and twist

Term: Renderer

Definition: Correspondence that assigns an Actual state to particles fields mass distance and time in Reality

Term: Reality

Definition: Effective reality layer calculated by observers with standard physical equations

Term: Record

Definition: Record preserving observations sources uncertainties and scope of application

Term: Ledger

Definition: Ledger managing module versions ownership updates and failure history

Term: Common carrier

Definition: Common basis enabling change and transmission in Actual rather than an additional particle

Term: Phenotype

Definition: Result rendered as an observable property through Renderer

Term: Twist

Definition: Residual global or local relation left after a state is transported around a closed connection

Term: Mass phenotype

Definition: State in which local twist stress is assigned as an additional gravitational source in Reality

Term: Relational opening

Definition: Actual interpretation of the Big Bang in which distinguishable states paths and quantization first become established

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